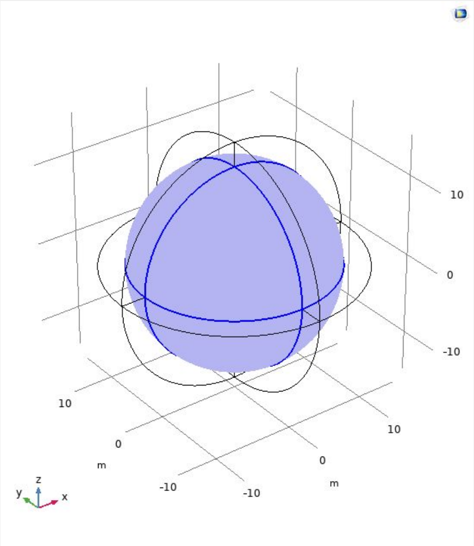
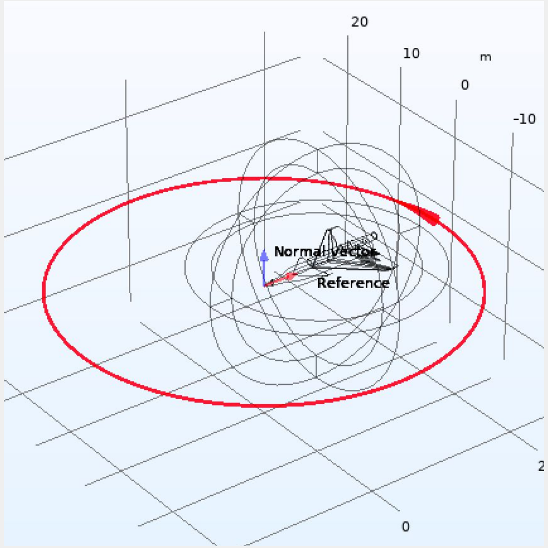
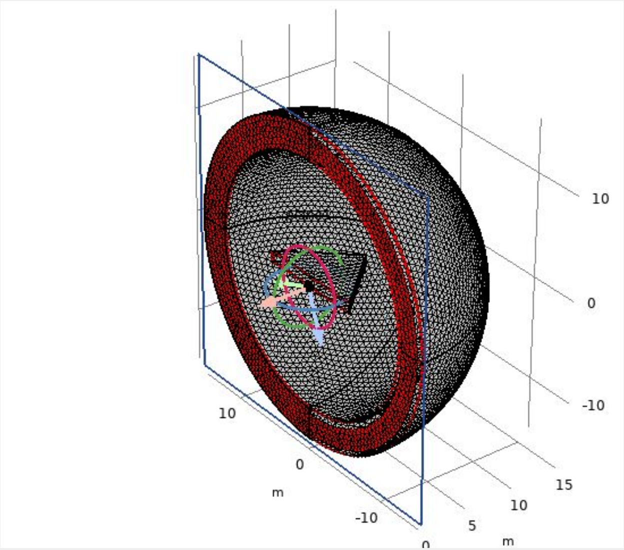
RADAR field physics initialization





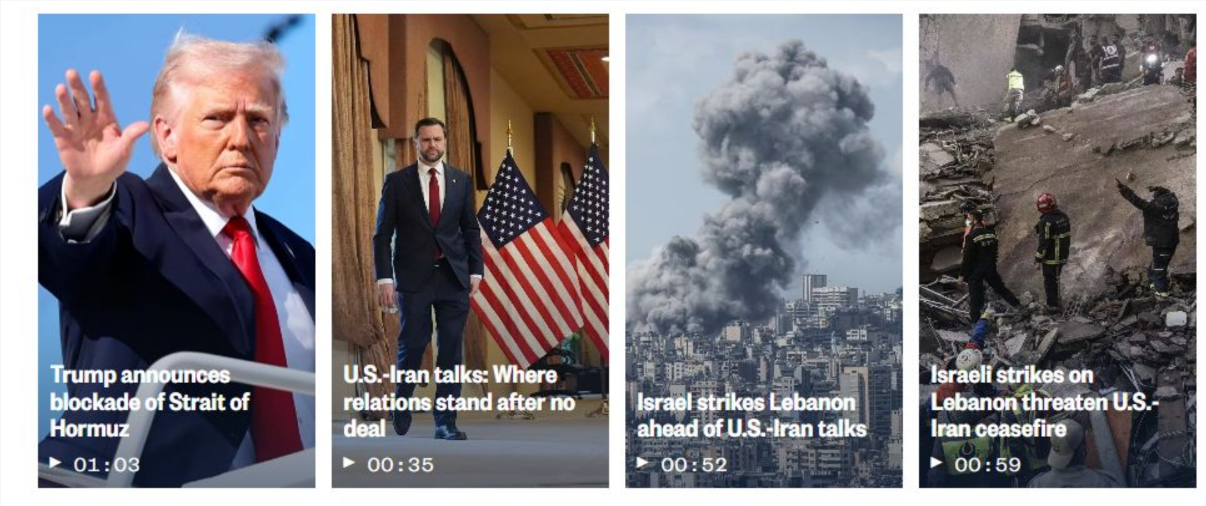


<u>Far-field domain</u>

At sufficient distances, a point source of radio waves can be characterized as a plane wave

<u>Impedance boundary condition</u>

At 100MHz, the skin depth of aluminum is on the order of 1e-6, much less than its thickness



13 April, 2026

https://www.nbcnews.com/middle-east-conflict

2 March, 2022

https://www.newyorker.com/news/annals-of-communications/a-mediated-view-of-the-war-in-ukraine

At the 2025 NATO Summit in The Hague, Allies made a commitment to investing **5% of Gross Domestic Product** (GDP) annually on core defence requirements and defence- and security-related spending by 2035.

[Defense expenditure and NATO's 5% commitment]

Canada is investing more than **$63 billion** in defence across the Department of National Defence (DND), the Canadian Armed Forces (CAF), and other government partners including military personnel, readiness, equipment, and infrastructure.

[Canada achieves the 2% of gross domestic product defense spending benchmark]

Russo-Ukrainian war

Canada has committed over **$25.5 billion** in multifaceted assistance for Ukraine .

Economic, humanitarian and development assistance, and security and stabilization support – Russia's invasion of Ukraine

Israeli-Palestinian conflict

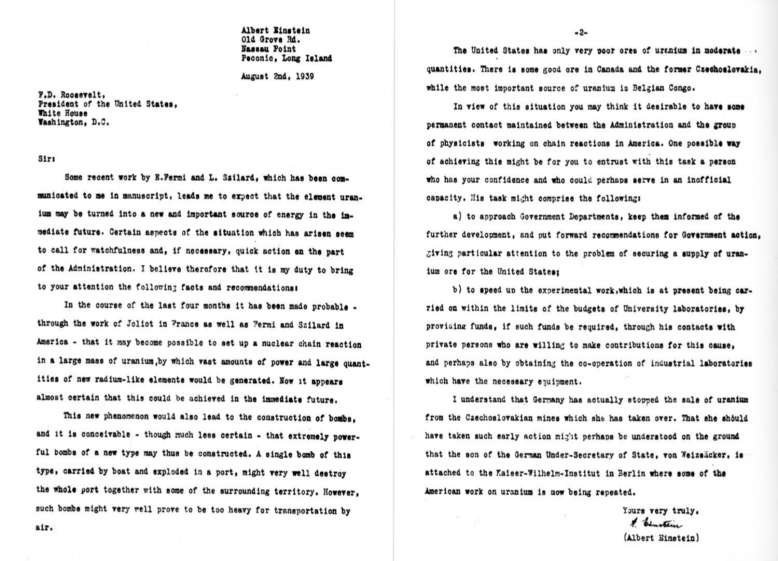
Since the current conflict began, and with the additional resources announced in July 2025, Canada's contributions now total more than **$355 million.**

Backgrounder: Canada's International Assistance to the West Bank and Gaza

Israeli-Syrian conflict (during Syrian Civil War)

Since 2016, Canada has committed more than **$4.7 billion** in funding for Syria and countries hosting Syrian refugee populations.

Backgrounder: Canada announces measures and support for the people of Syria



Albert Einstein
Old Grove Rd.
Nassau Point
Peconic, Long Island

August 2nd, 1939

F.D. Roosevelt,
President of the United States,
White House
Washington, D.C.

Sir:

Some recent work by E.Fermi and L. Szilard, which has been communicated to me in manuscript, leads me to expect that the element uranium may be turned into a new and important source of energy in the immediate future. Certain aspects of the situation which has arisen seem to call for watchfulness and, if necessary, quick action on the part of the Administration. I believe therefore that it is my duty to bring to your attention the following facts and recommendations:

In the course of the last four months it has been made probable - through the work of Joliot in France as well as Fermi and Szilard in America - that it may become possible to set up a nuclear chain reaction in a large mass of uranium,by which vast amounts of power and large quantities of new radium-like elements would be generated. Now it appears almost certain that this could be achieved in the immediate future.

This new phenomenon would also lead to the construction of bombs, and it is conceivable - though much less certain - that extremely powerful bombs of a new type may thus be constructed. A single bomb of this type, carried by boat and exploded in a port, might very well destroy the whole port together with some of the surrounding territory. However, such bombs might very well prove to be too heavy for transportation by air.

-2-

The United States has only very poor ores of uranium in moderate quantities. There is some good ore in Canada and the former Czechoslovakia, while the most important source of uranium is Belgian Congo.

In view of this situation you may think it desirable to have some permanent contact maintained between the Administration and the group of physicists working on chain reactions in America. One possible way of achieving this might be for you to entrust with this task a person who has your confidence and who could perhaps serve in an inofficial capacity. His task might comprise the following:

a) to approach Government Departments, keep them informed of the further development, and put forward recommendations for Government action, giving particular attention to the problem of securing a supply of uranium ore for the United States;

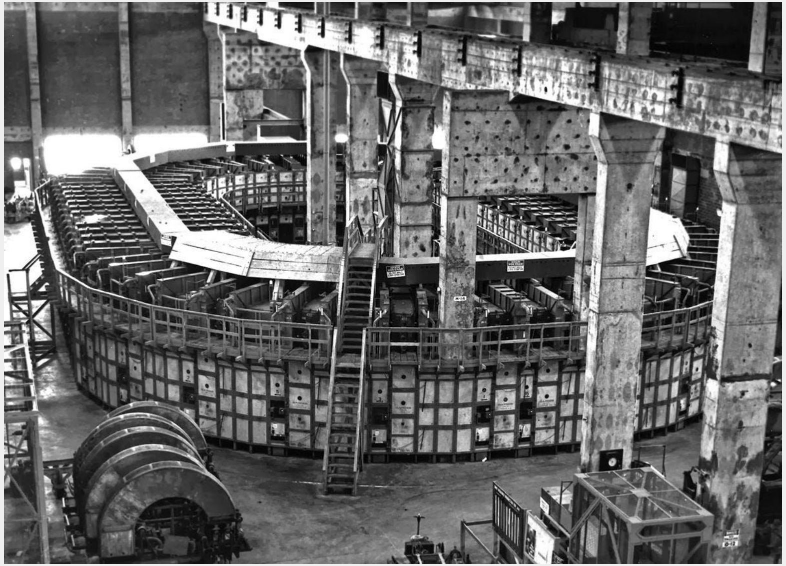
b) to speed up the experimental work,which is at present being carried on within the limits of the budgets of University laboratories, by providing funds, if such funds be required, through his contacts with private persons who are willing to make contributions for this cause, and perhaps also by obtaining the co-operation of industrial laboratories which have the necessary equipment.

I understand that Germany has actually stopped the sale of uranium from the Czechoslovakian mines which she has taken over. That she should have taken such early action might perhaps be understood on the ground that the son of the German Under-Secretary of State, von Weizsäcker, is attached to the Kaiser-Wilhelm-Institut in Berlin where some of the American work on uranium is now being repeated.

Yours very truly,
A. Einstein
(Albert Einstein)

1939 letter by Einstein and Szilard urging Roosevelt to develop a nuclear program.

https://en.wikipedia.org/wiki/Einstein%E2%80%93Szilard_letter



State-sponsored Manhattan Project to develop the atomic bomb

https://www.construction-physics.com/p/an-engineering-history-of-the-manhattan

Stealth

Arctic fox camouflaged in snow

https://manilastandard.net/?p=314271425

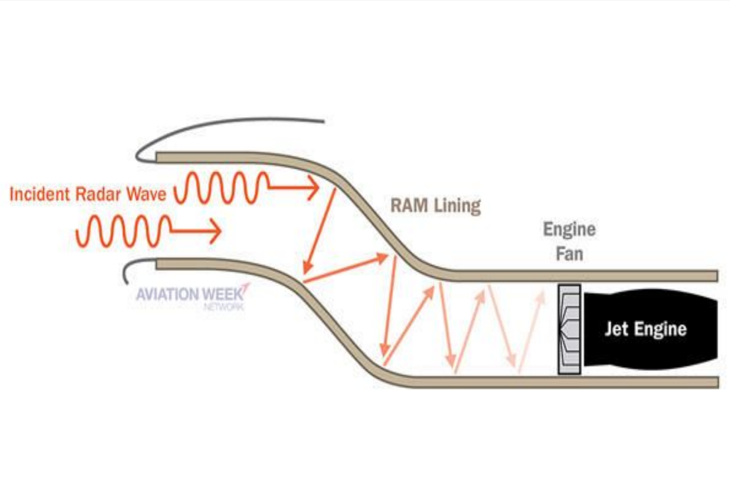
Ninja in black clothing

https://www.historyskills.com/classroom/year-8/ninjas/?srsltid=AfmBOoo_xmGpO9fpdvBlTijDDLBakLC7qEZB6dqg7Rh2JGSPz5x_UQtj

Ground squirrel chewing on snake skin to mask its scent

https://twig.technology/blog/ground-squirrel-rattlesnake-deterrence

Factors of stealth seen in modern aircraft technology

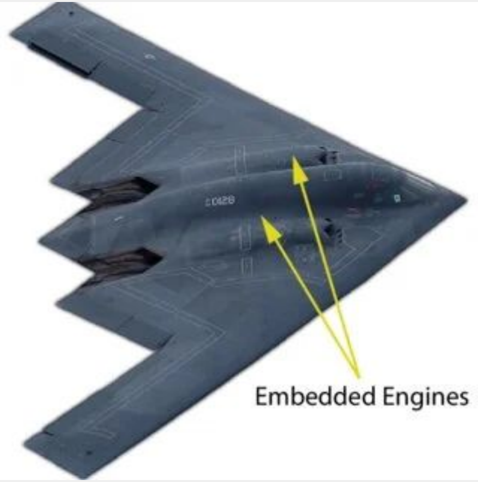


Radiation-absorbent materials

https://aviationweek.com/aerospace/aircraft-propulsion/magic-behind-radar-absorbing-materials-stealthy-aircraft

Camouflage

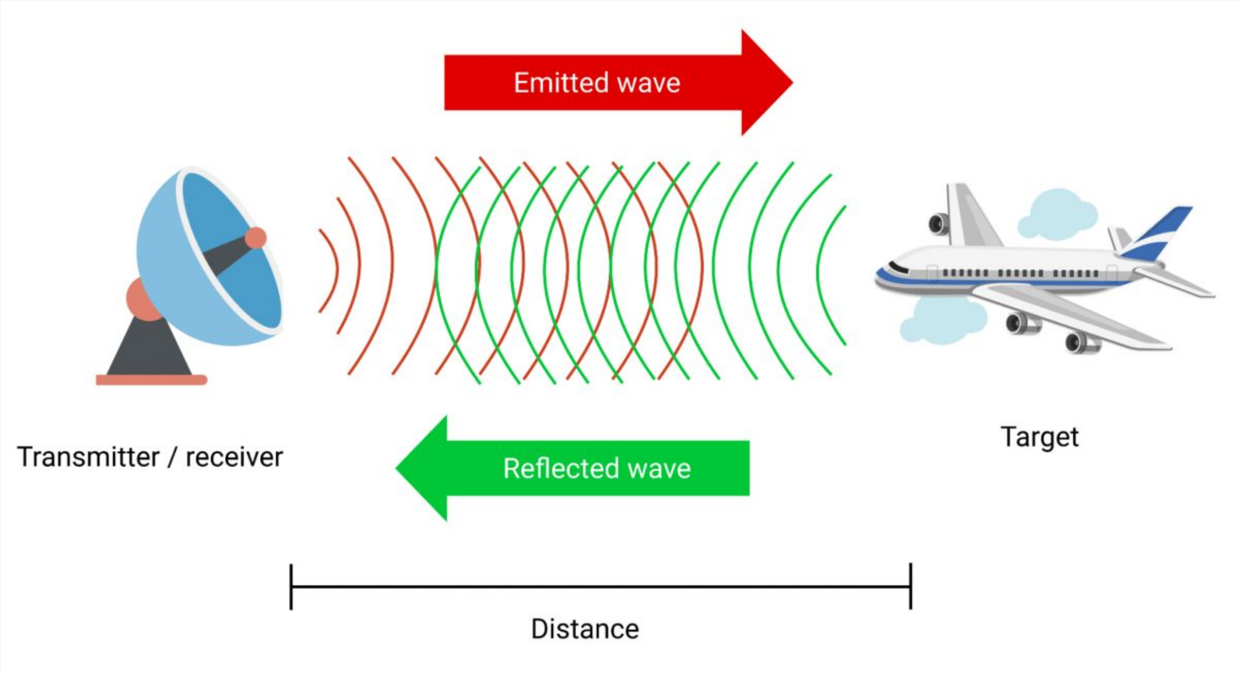
https://www.defencexp.com/stealth-technology-how-aircraft-becomes-invisible/



Low acoustic signature

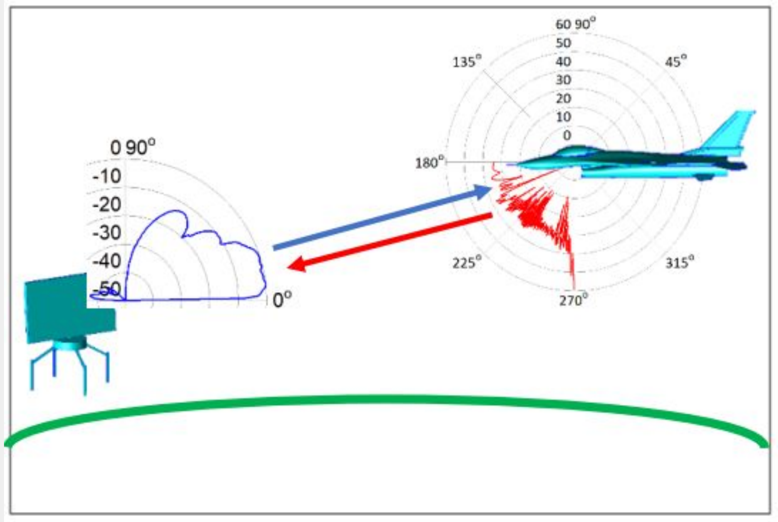
https://www.researchgate.net/figure/B-2-Spirit-stealth-bomber-with-embedded-engine-configuration_fig1_268574716

RADAR technology



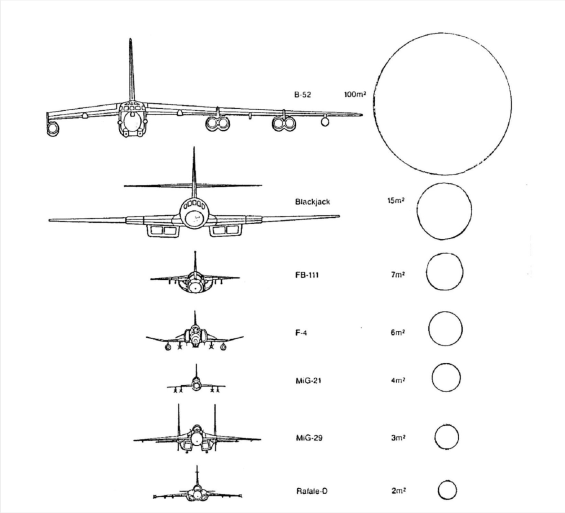
https://www.thinkautonomous.ai/blog/how-radars-work/

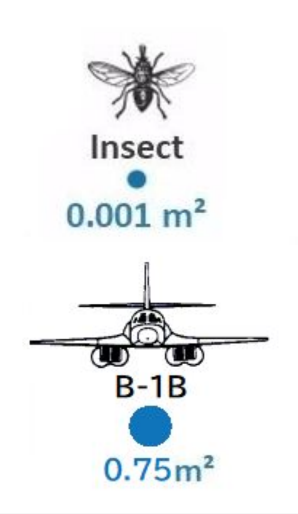
Radio Cross Section (RCS)



Ground-based radar and aircraft monostatic scattering.

https://wipl-d.com/wp-content/uploads/2021/06/RCS_016_WIPL-D_Monostatic_Bistatic_Multistatic_Radar_Scenarios_2021.pdf

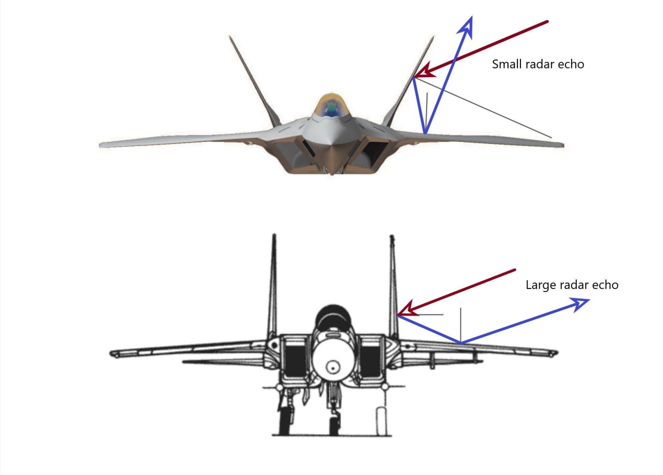


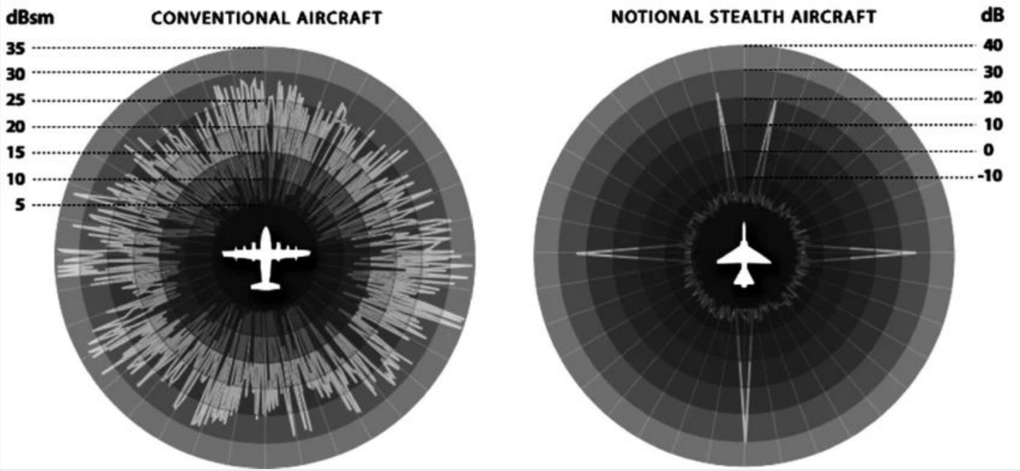


Comparison of RCS of various objects in flight

https://www.slideshare.net/slideshow/report-on-stealth-technology/65096843

Radio signature and back-scattering





https://medium.com/geekculture/heres-how-a-2-billion-dollar-flying-wing-can-disappear-from-the-radar-screen-a8e0b9d6cd1a

https://www.researchgate.net/figure/Illustration-of-a-Conventional-and-b-Stealth-Aircraft-Radar-Cross-Section-Signature_fig4_308781073

Variance in shape across stealth aircrafts

Northrop B-2 Spirit

https://en.wikipedia.org/wiki/Northrop_B-2_Spirit

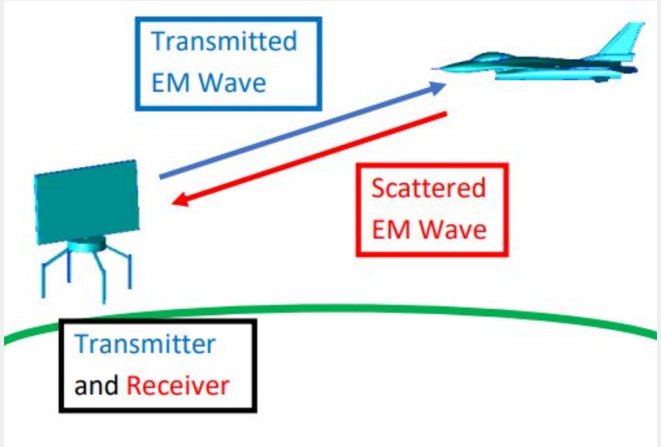
Lockheed Martin F-35 Lightning II

https://en.wikipedia.org/wiki/Lockheed_Martin_F-35_Lightning_II

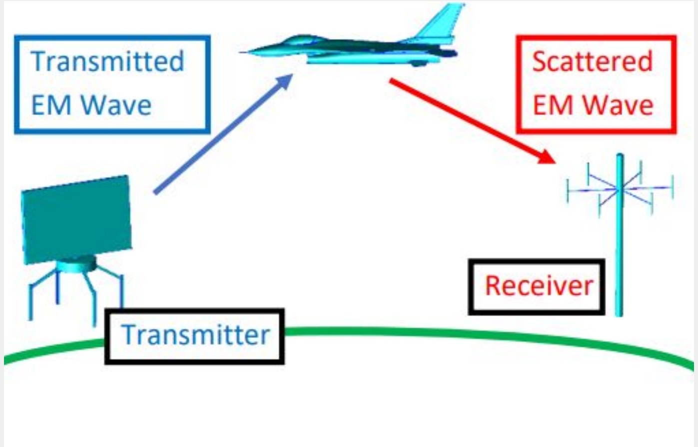
Lockheed F-117 Nighthawk

https://en.wikipedia.org/wiki/Lockheed_F-117_Nighthawk

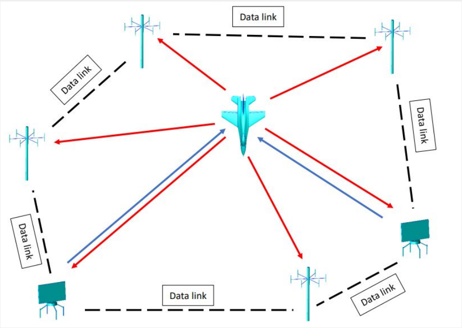
Consideration: types of radio wave detection



Monostatic scattering



Bistatic scattering



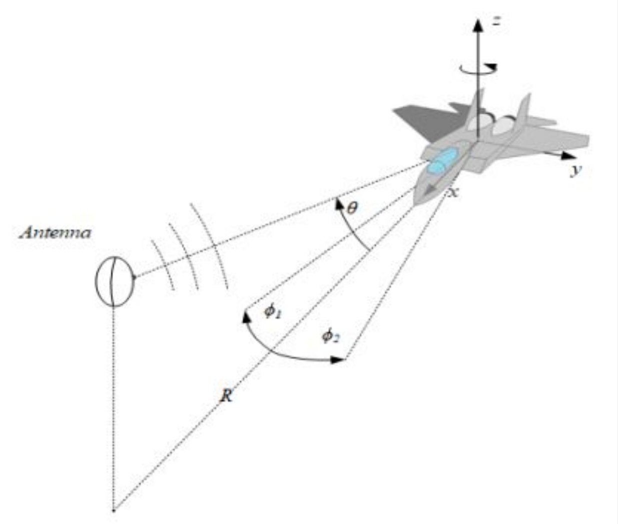
Multistatic scattering??

https://wipl-d.com/wp-content/uploads/2021/06/RCS_016_WIPL-D_Monostatic_Bistatic_Multistatic_Radar_Scenarios_2021.pdf

Consideration: orientation

Cockpit of B-2 stealth plane

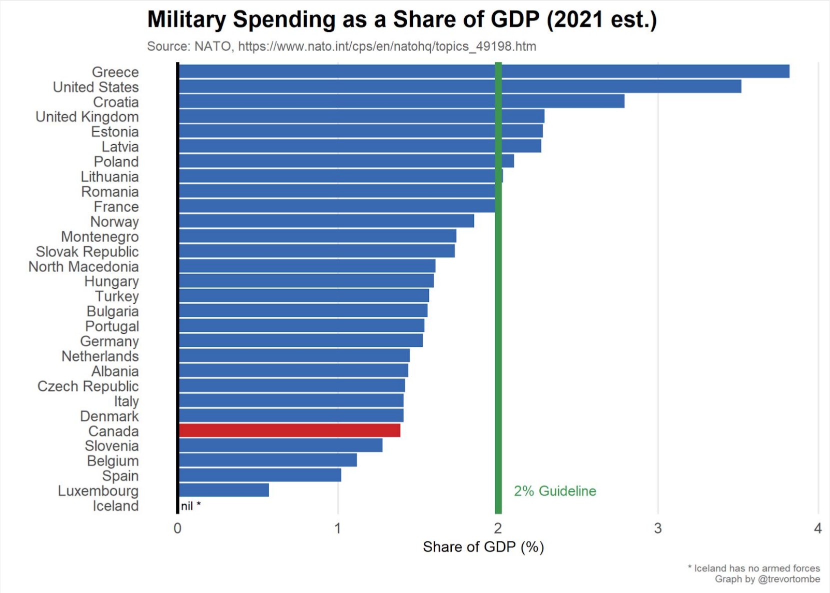
https://theaviationist.com/2019/04/29/here-are-some-interesting-details-weve-noticed-in-the-first-ever-video-filmed-inside-a-b-2s-cockpit-while-in-flight/



Point of earliest detection

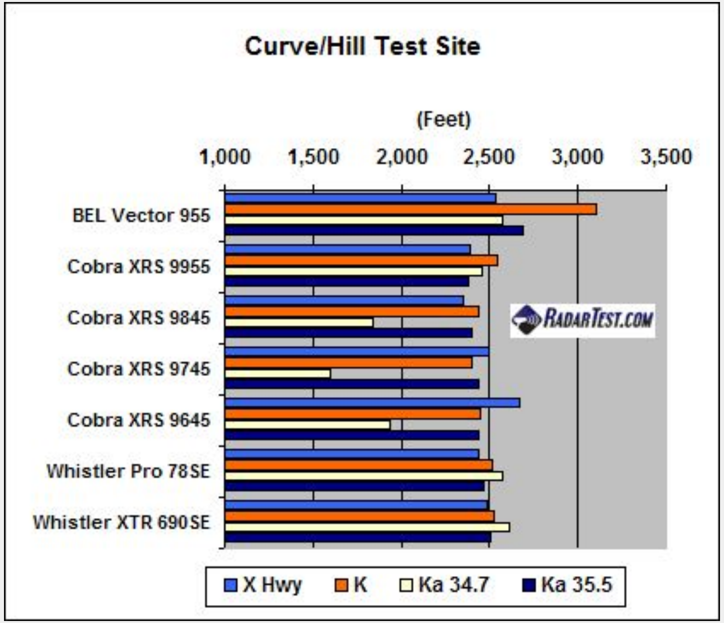
https://wipl-d.com/wp-content/uploads/2018/09/RCS_007_WIPL-D_Detecting_Aircraft_Shape_via_SAR.pdf

Proposal: (physics model capable of) optimizing geometry of an aircraft body to minimize its RADAR visibility



Canada has increased military spending

https://thehub.ca/2022/03/30/what-increased-military-spending-may-mean-for-canadas-budget/

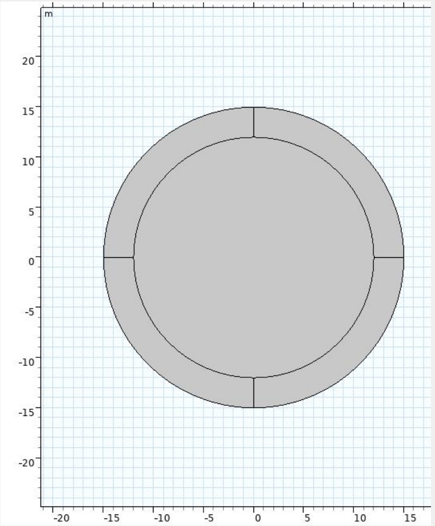


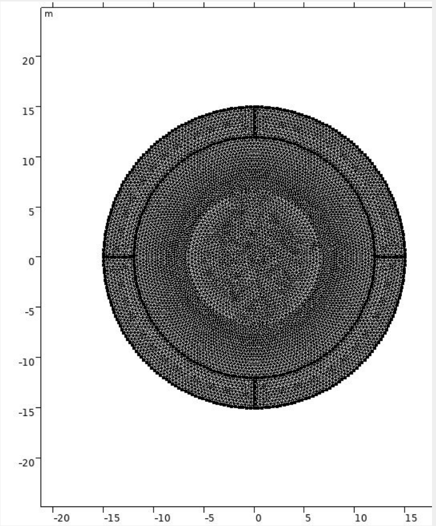
RADAR detectors fulfil Moore's Law

https://radartest.com/article.asp?articleid=100623

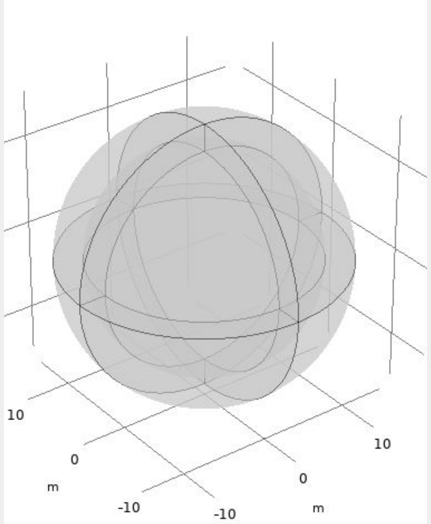
COMSOL RADAR field model

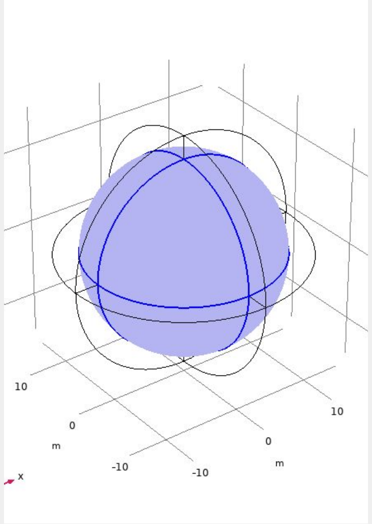
2D





3D





Physics: *Electromagnetic waves, frequency domain*

$$\boxed{\begin{gathered}\mathbf{E}_{\mathrm{b,2D}} = \exp\left(jk_0\big(xcos(\phi) + ysin(\phi)\big)\right)\boldsymbol{e}_z \\ \mathbf{E}_{\mathrm{b,3D}} = \exp(jk_0 x)\,\boldsymbol{e}_z\end{gathered}}$$

j is the imaginary unit

$k_0 = 2\pi f/c$ is the wave number in vacuum

$c = 3 * 10^8$ m/s is the speed of light

$f = 100$ MHz is the frequency

ϕ is the angle of incidence

$\phi = 0$ corresponds to a wave incident from the positive x direction (and hence propogating in the negative x direction, from right to left in the model

$\mathbf{E}_{\mathrm{rel}} = \mathbf{E} - \mathbf{E}_{\mathrm{b}}$ where $\boldsymbol{E}$ is the total, measurable field.

$$\boxed{\nabla \times (\mu_r^{-1}\nabla \times \mathbf{E}_{\mathrm{rel}}) - \left(\epsilon_r - j\frac{\sigma}{\omega\epsilon_0}\right)k_0^2\mathbf{E}_{\mathrm{rel}} = 0}$$

2D and 3D wave equation

Scatter field formulation

Background electric field

https://doc.comsol.com/6.3/doc/com.comsol.help.rf/RFModuleUsersGuide.pdf

Physics: *Electromagnetic waves, frequency domain*

The relative field is the difference between the measured field caused by the presence of the plane and the background field. It is utilized to describe how detectable the boat is with the radar – its RCS. The RCS of 3D scatterer is defined as

$$\boxed{\sigma_{3D} = \lim_{\mathrm{r}\to\infty} 4\pi r^2 \frac{|\mathbf{E}_{\mathrm{rel}}|^2}{|\mathbf{E}_{\mathrm{b}}|^2}}$$

The RCS per unit length of a 2D scatterer addresses monostatic scattering characteristics at the angle where the incident wave comes from. It is defined as

$$\boxed{\sigma_{2D} = \lim_{\mathrm{r}\to\infty} 2\pi r \frac{|\mathbf{E}_{\mathrm{rel}}|^2}{|\mathbf{E}_{\mathrm{b}}|^2}}$$

2D and 3D RCS

https://doc.comsol.com/6.3/doc/com.comsol.help.rf/RFModuleUsersGuide.pdf

Physics: *Electromagnetic waves, frequency domain*

The far field can be calculated from the near field using the Stratton-Chu formula:

$$\mathbf{E}_{p,3D} = \frac{jk}{4\pi}\boldsymbol{r}_0 \times \int_S [\boldsymbol{n} \times \mathbf{E} - \eta \boldsymbol{r}_0 \times (\boldsymbol{n} \times \boldsymbol{H})] \exp\left(jk\boldsymbol{r} \cdot \boldsymbol{r}_0\right) dS$$

$$\mathbf{E}_{p,2D} = \sqrt{j\lambda}\frac{k}{4\pi}\boldsymbol{r}_0 \times \int_S [\boldsymbol{n} \times \mathbf{E} - \eta \boldsymbol{r}_0 \times (\boldsymbol{n} \times \boldsymbol{H})]\exp\left(jk\boldsymbol{r} \cdot \boldsymbol{r}_0\right) dS$$

$\boldsymbol{E}$ and $\boldsymbol{H}$ are the fields on the "aperture"

$\boldsymbol{r}_0$ is the unit vector pointing from the origin to the field point p

$\boldsymbol{n}$ is the unit normal to the surface S

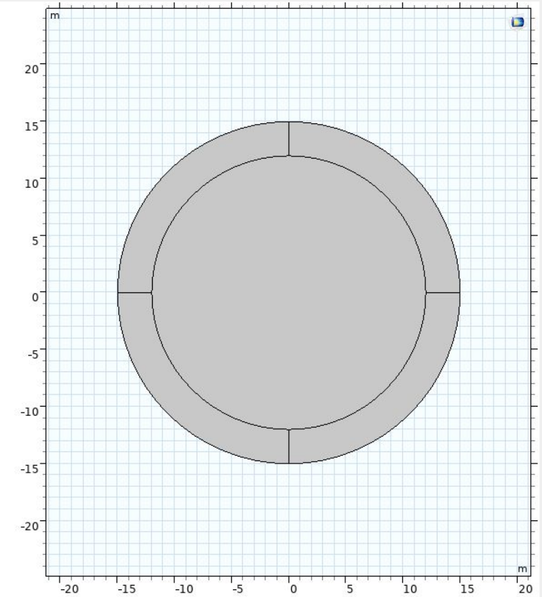
η is the impedance:

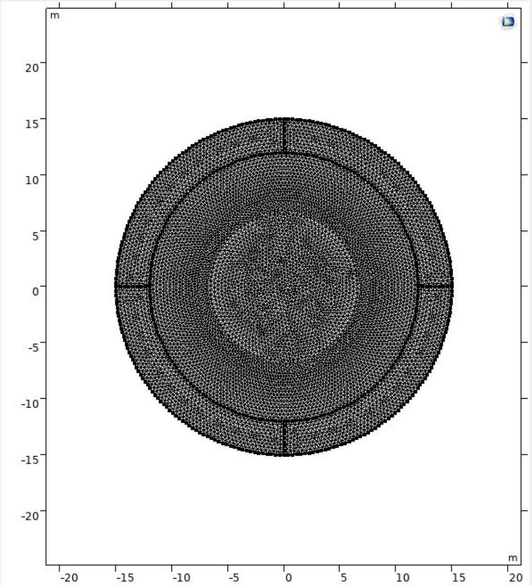
$$\eta = \sqrt{\frac{\mu}{\epsilon}}$$

Far field calculation

https://doc.comsol.com/6.3/doc/com.comsol.help.rf/RFModuleUsersGuide.pdf

2D RADAR field geometry in COMSOL

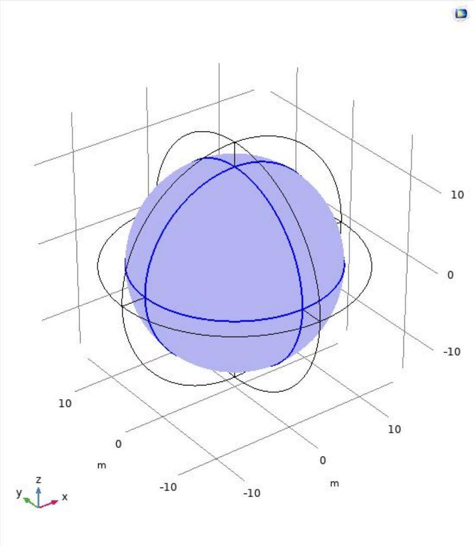


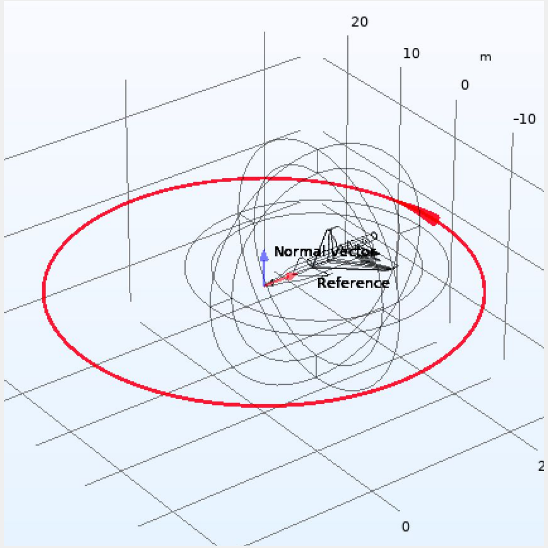


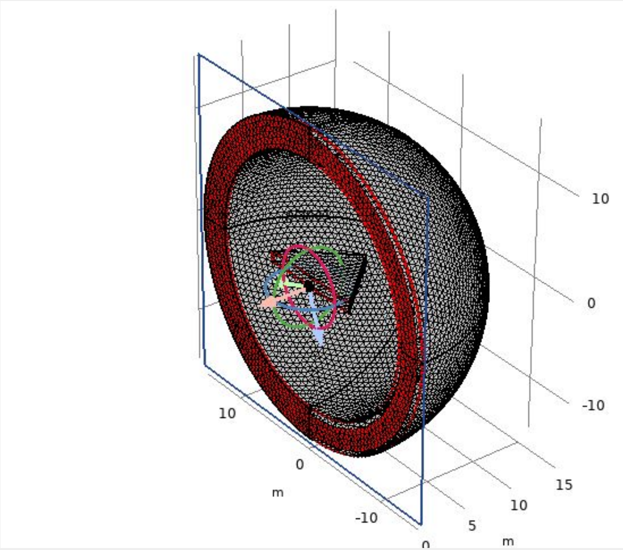
Inner circle: air surrounding the plane.

Outer circle: perfectly matched layer (PML), which minimizes unphysical reflections of the scattered wave as it leaves the model domain.

RADAR field physics initialization







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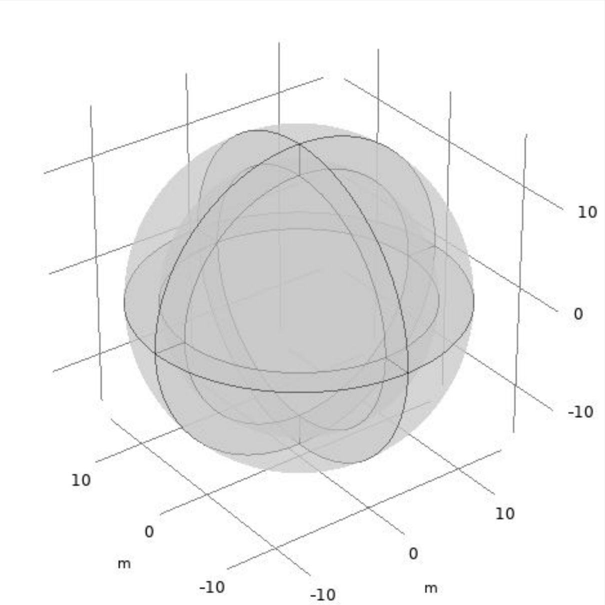
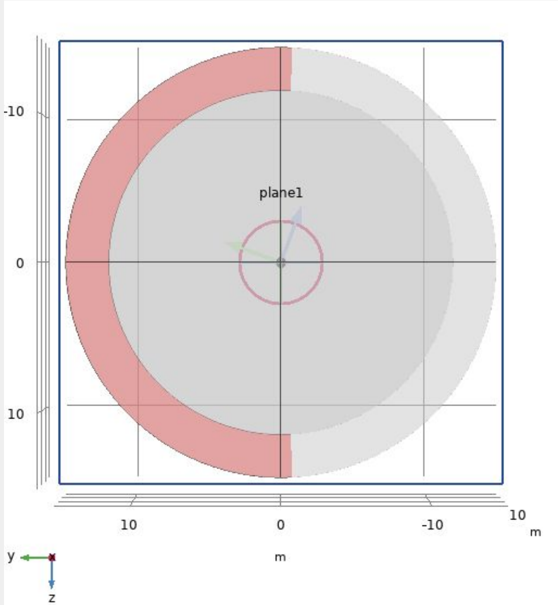
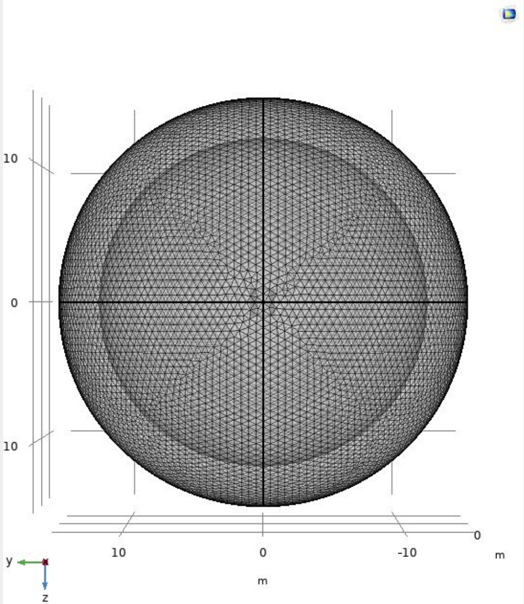
At 100MHz, the skin depth of aluminum is on the order of 1e-6, much less than its thickness

3D RADAR field geometry in COMSOL



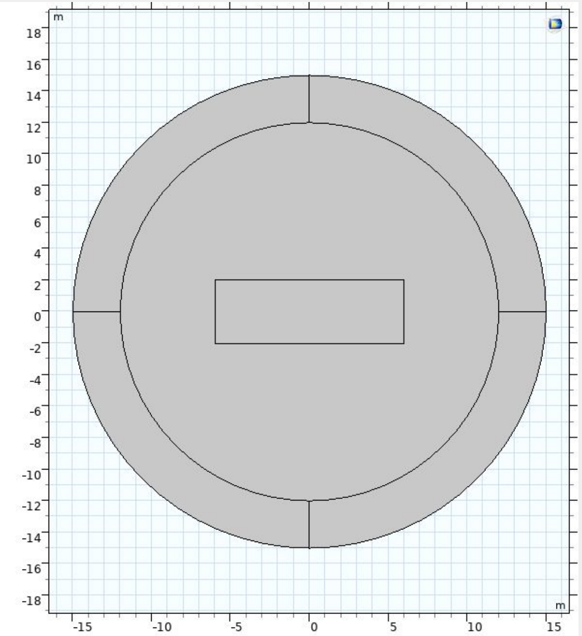


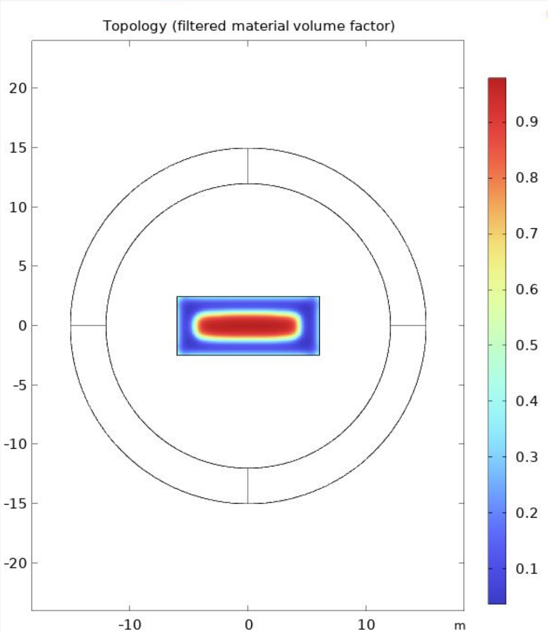


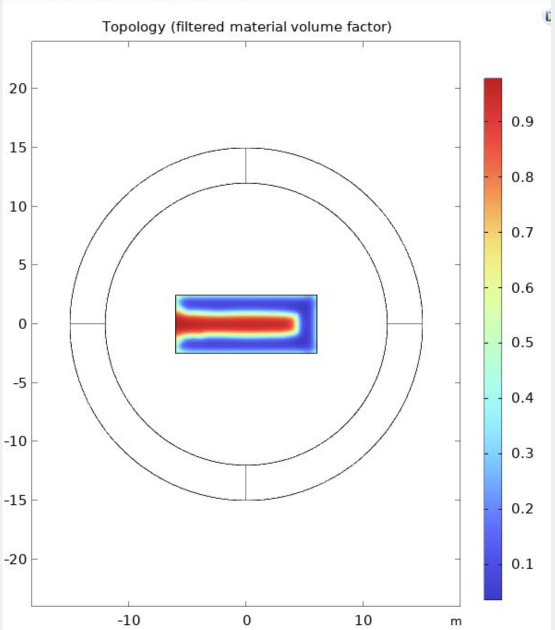
Inner sphere: air surrounding the plane.

Outer shell: perfectly matched layer (PML), which minimizes unphysical reflections of the scattered wave as it leaves the model domain.

Method 1: topology optimization





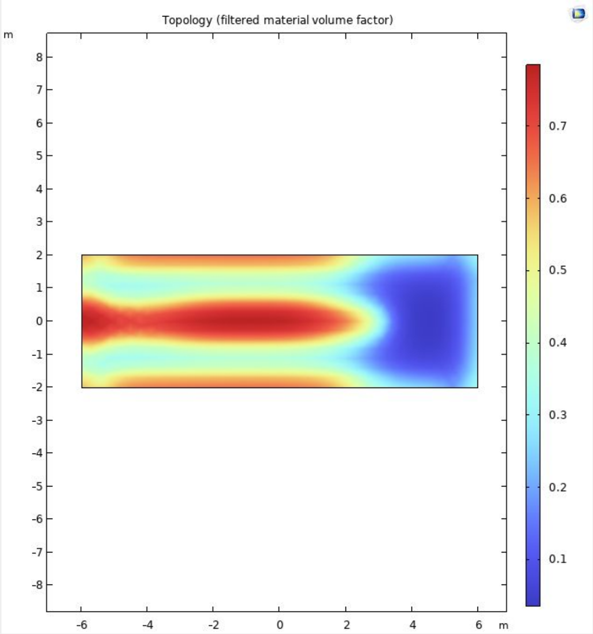


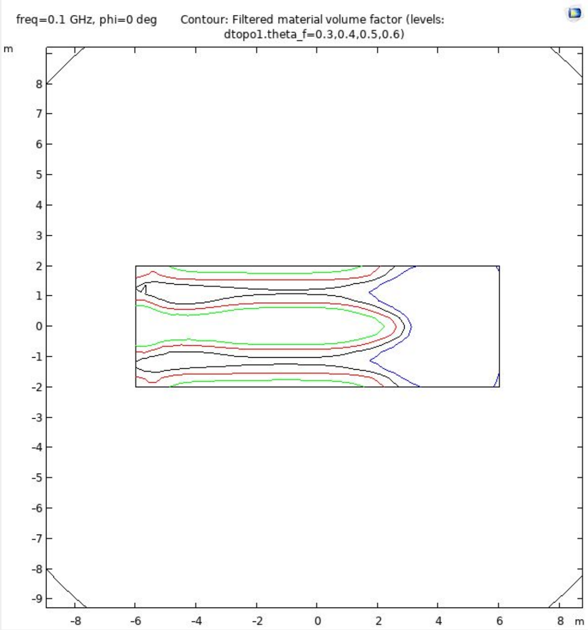
Constraints: total area

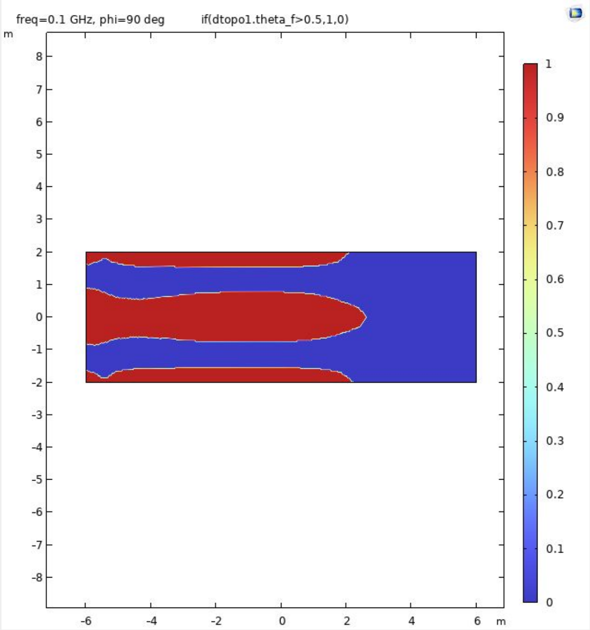
+ directional weighting

Method 1: topology optimization

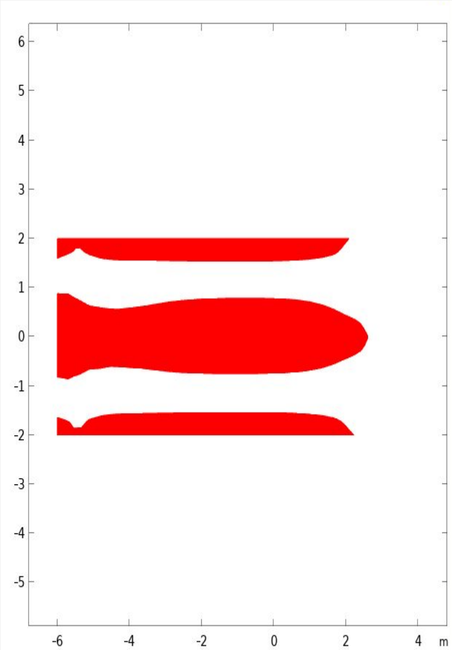
Optimal density arrangement → geometric part

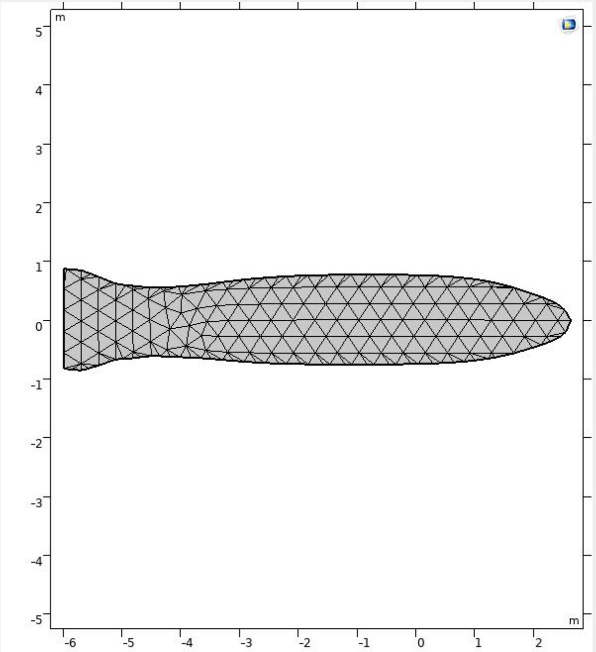


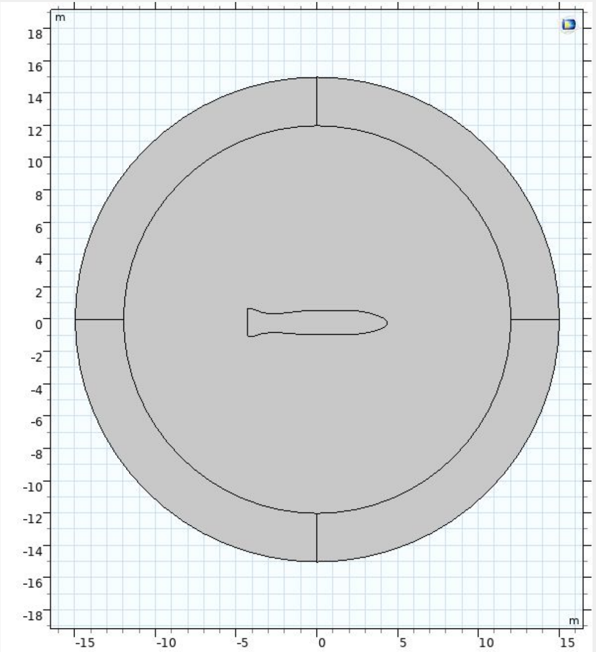




Method 1: topology optimization

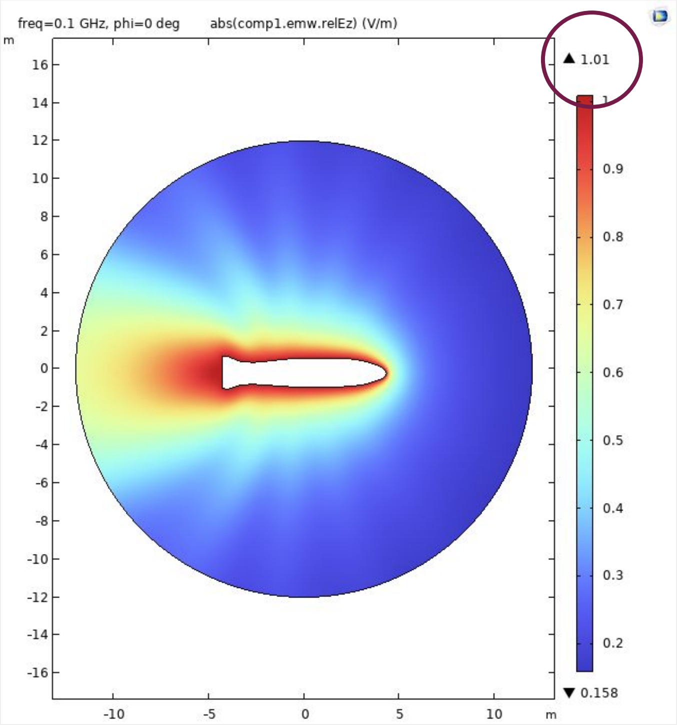




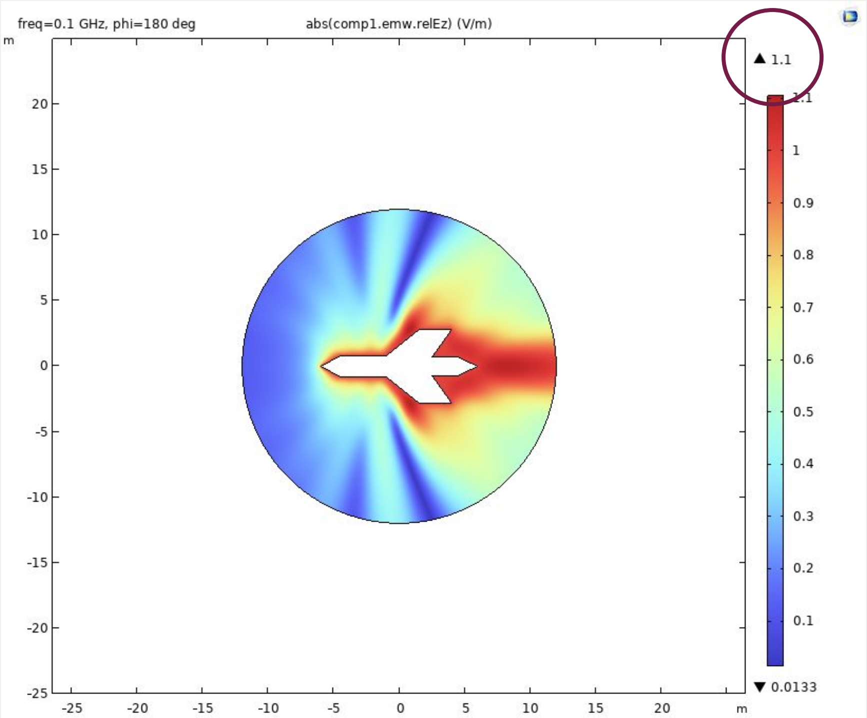


Method 1: topology optimization

2D results: relative electric field



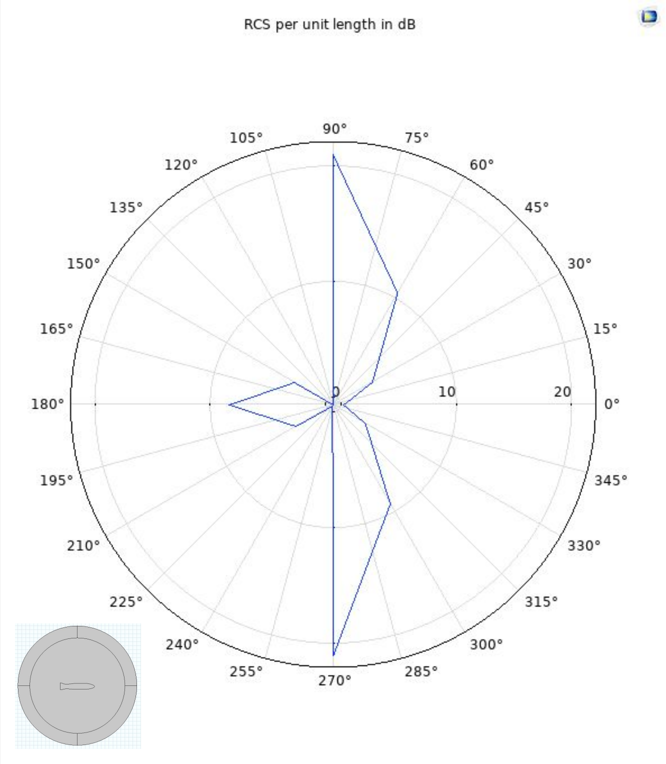
Compare: (reflected)

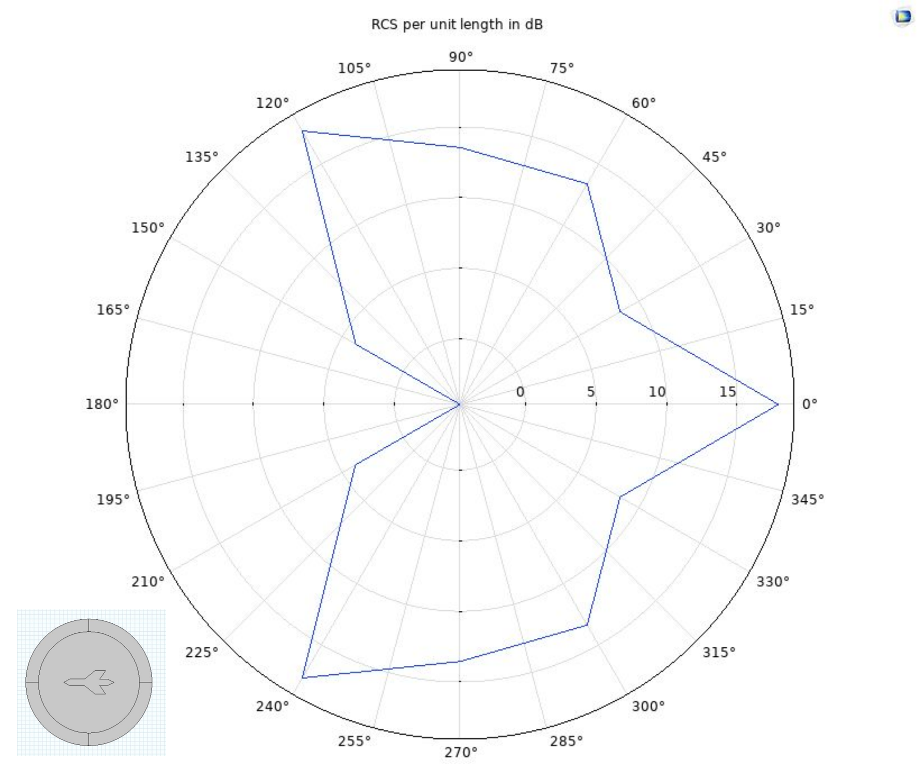


Method 1: topology optimization

2D results: RCS per unit length

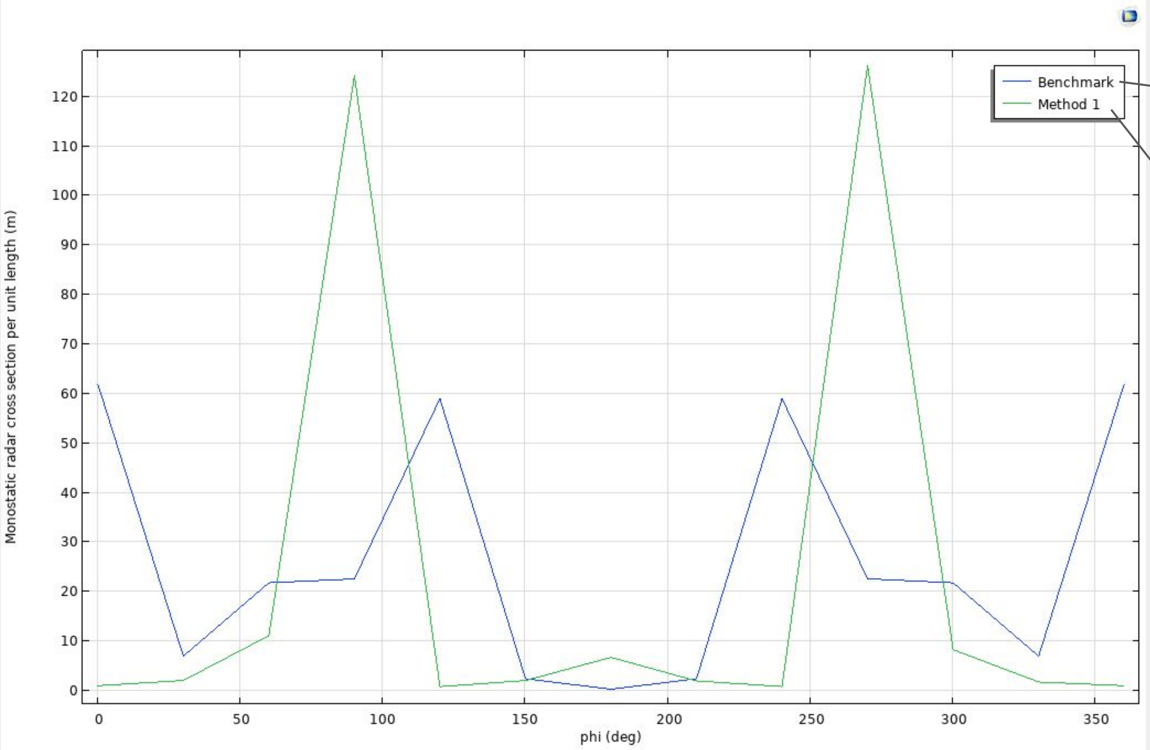
Compare: (reflected)

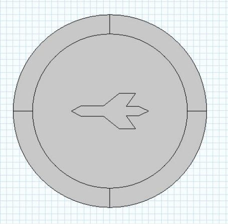


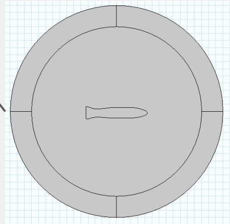


Method 1: topology optimization

2D results: RCS minimization

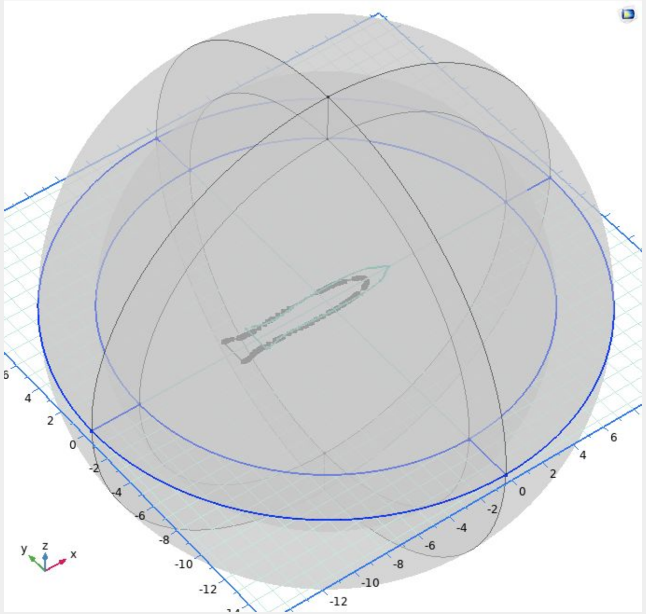


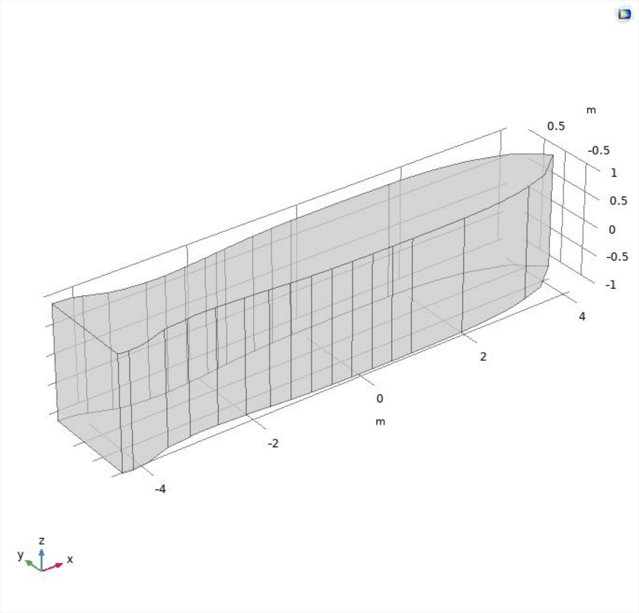


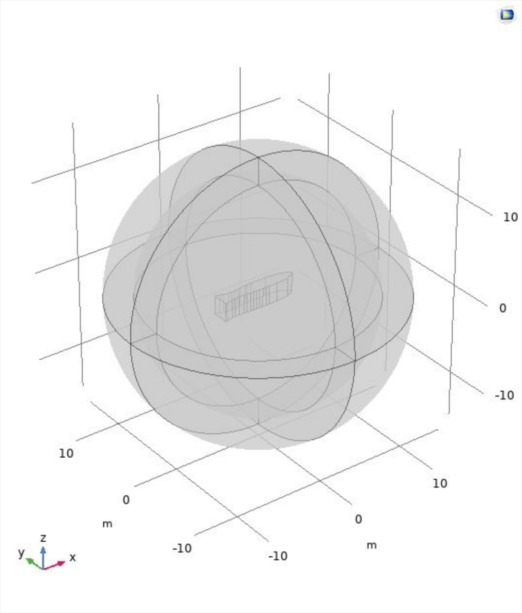


Method 1: topology optimization

2D part → 3D part

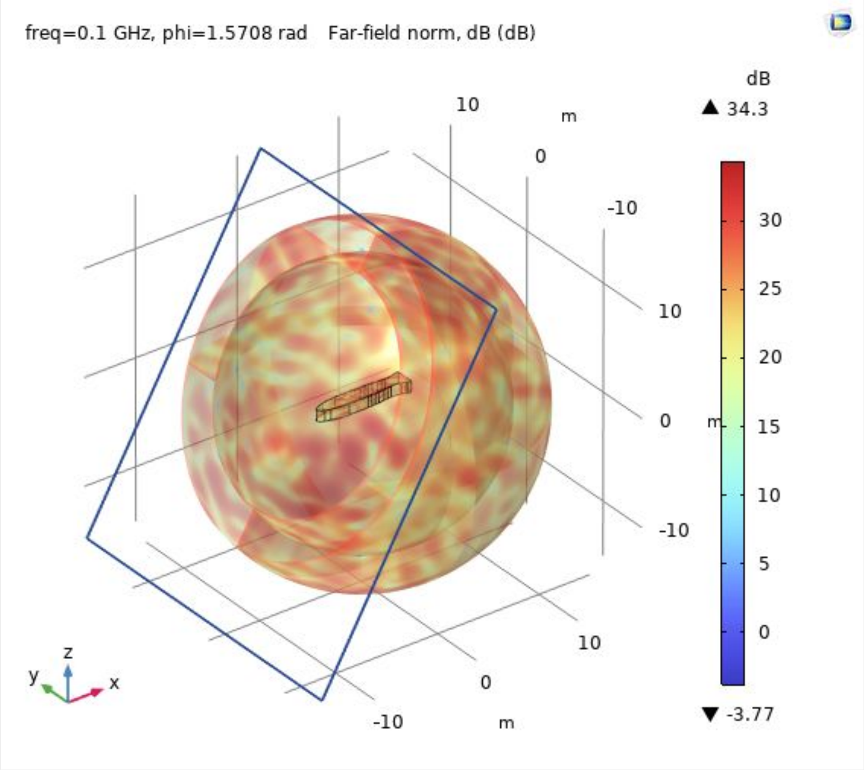




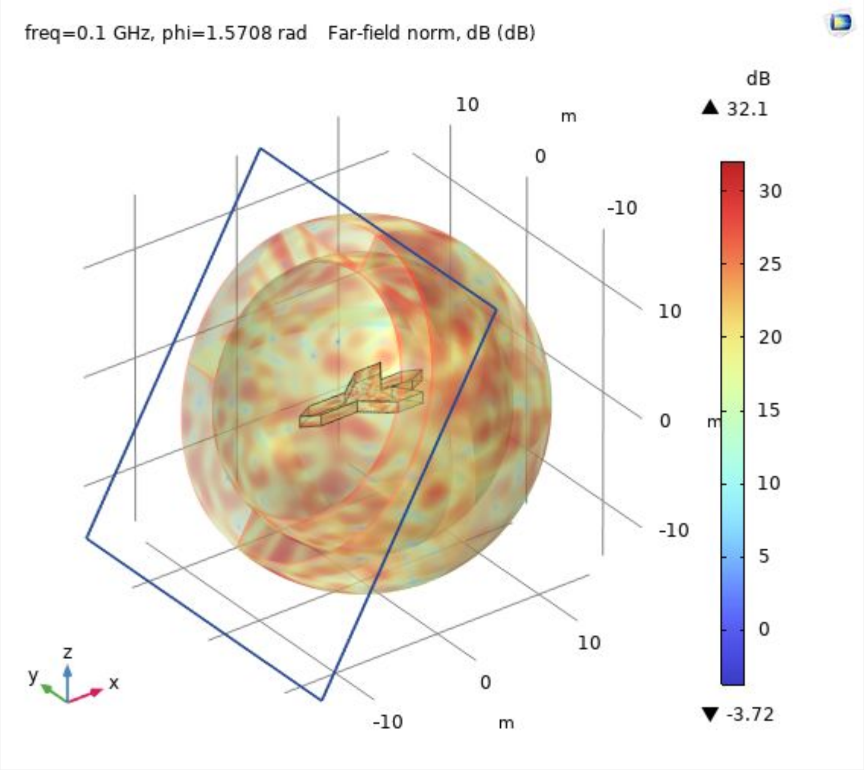


Method 1: topology optimization

Results: far-field norm

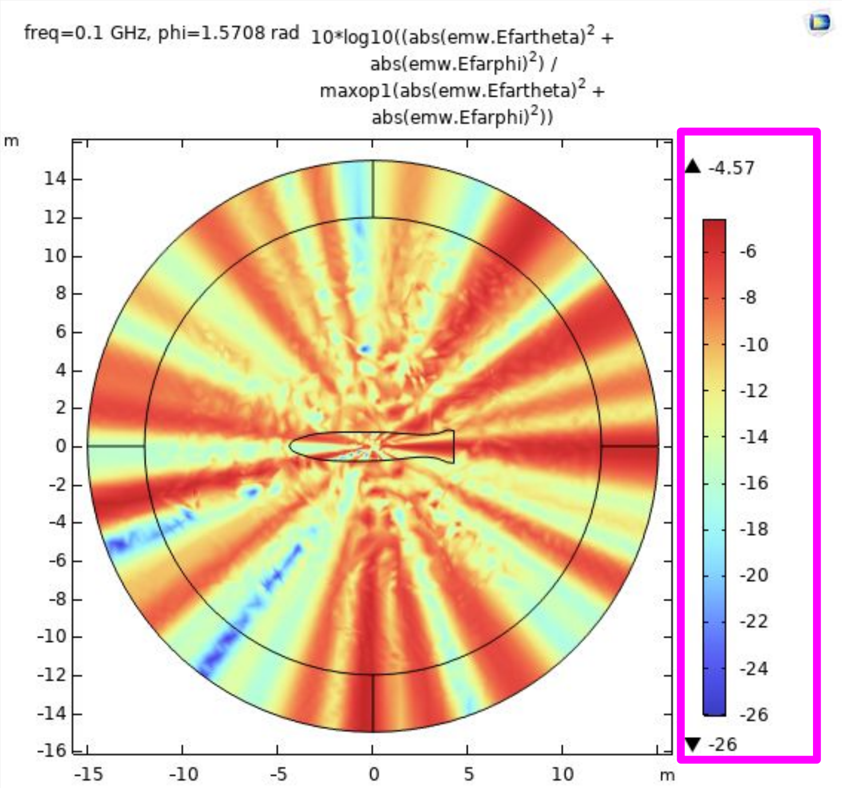


Compare:

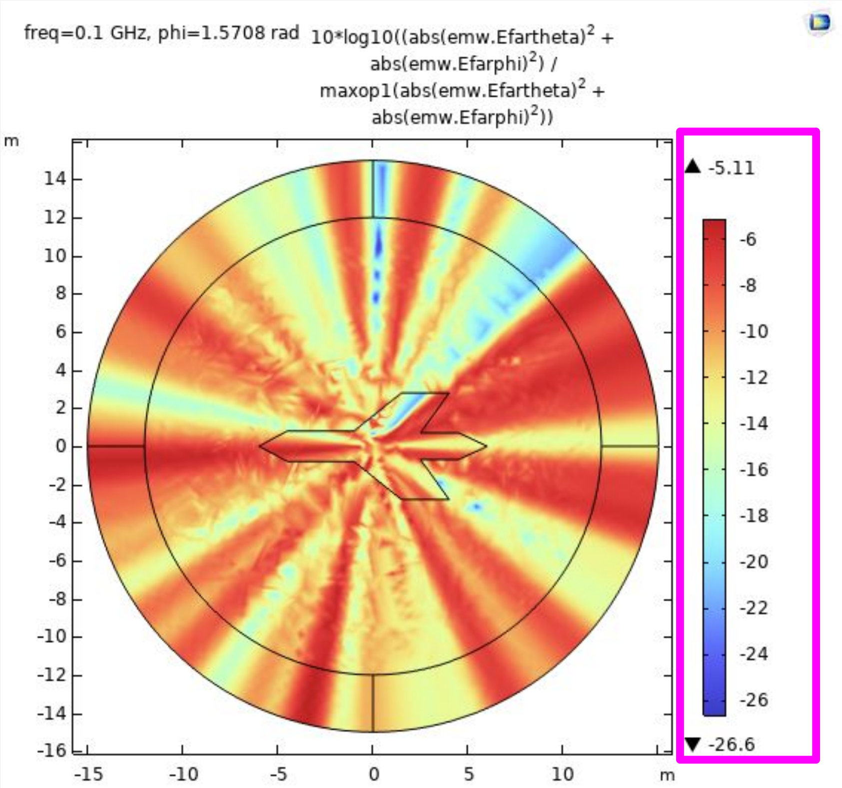


Method 1: topology optimization

Results: relative electric field



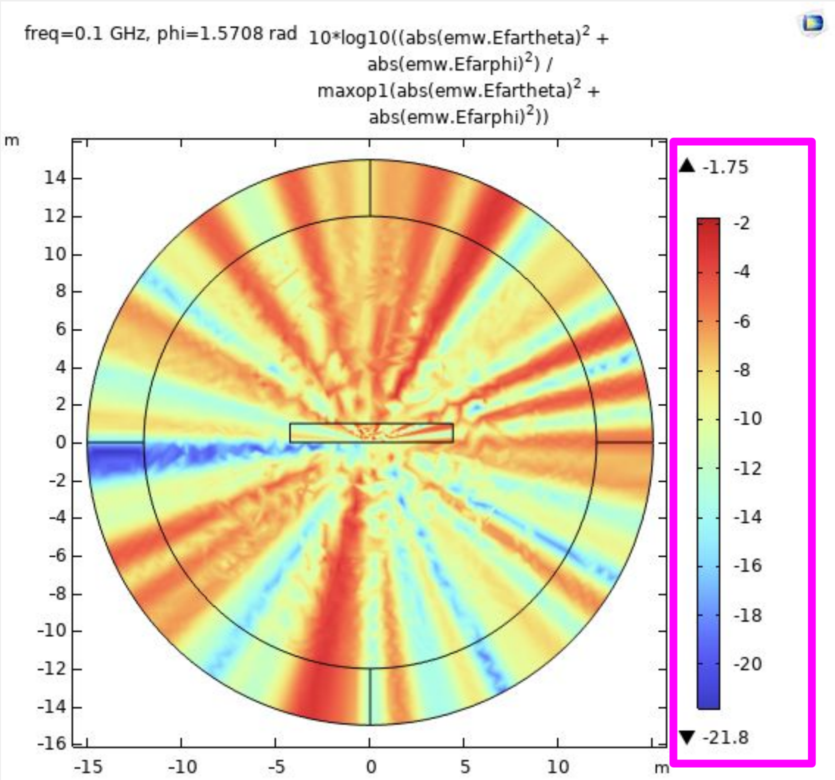
Compare:

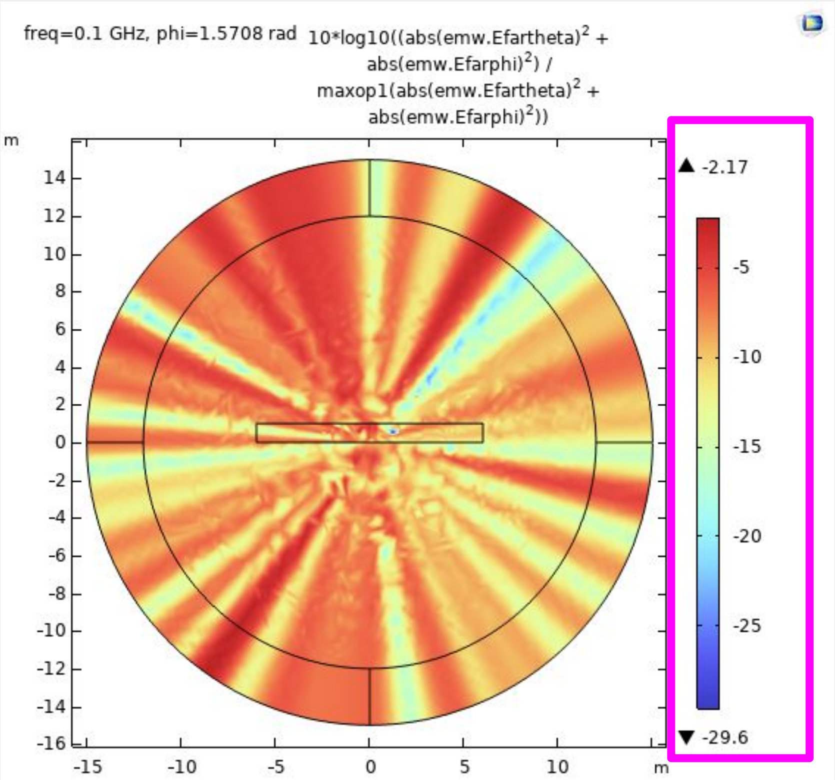


Method 1: topology optimization

Results: relative electric field

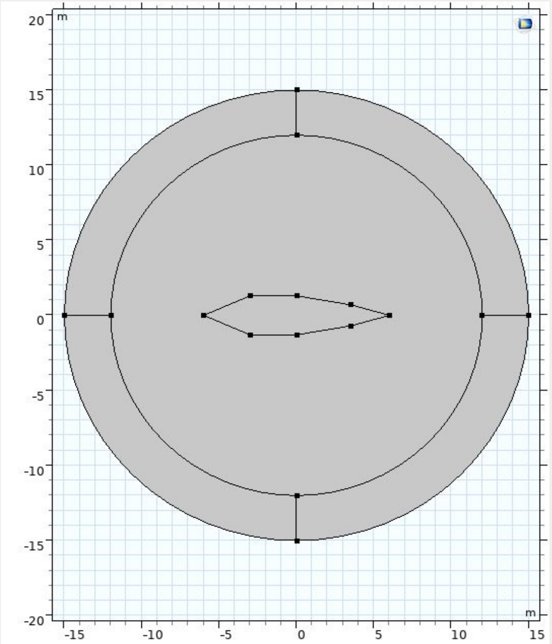
Compare:

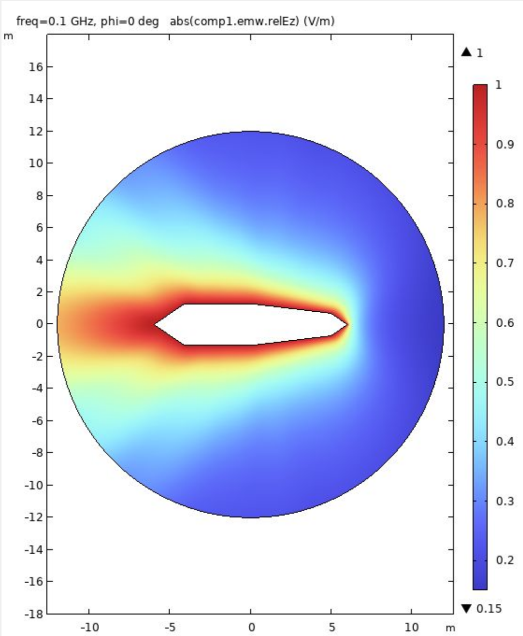


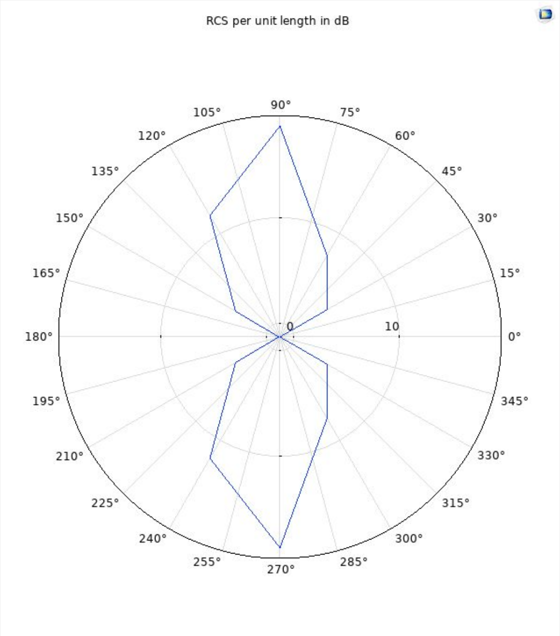


Method 2: parametric optimization

Variance in original shape





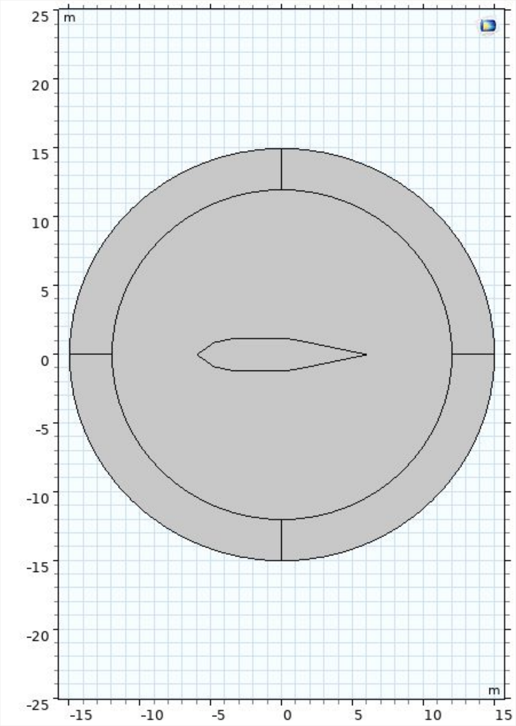


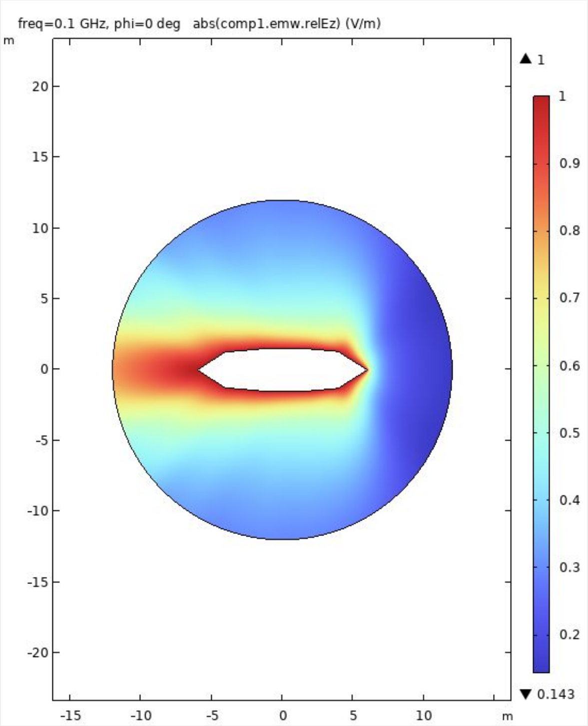
Controlled parameters: side length, interior angle.

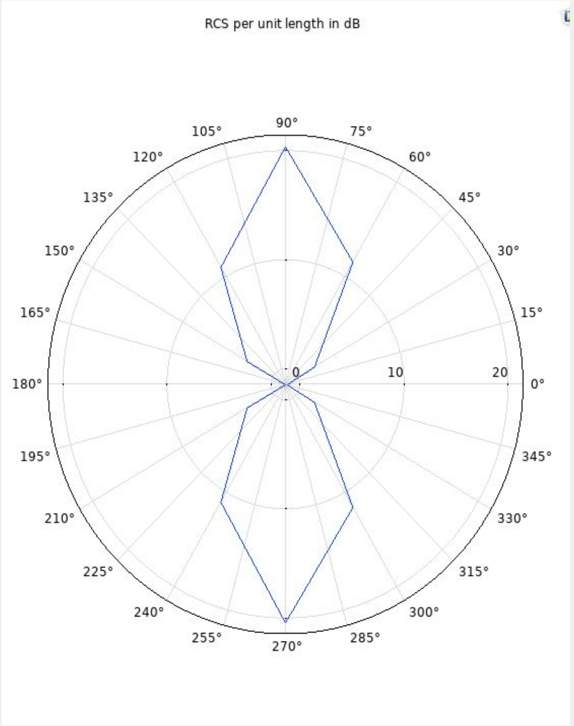
**Some restrictions on component size

Method 2: parametric optimization

Variance in original shape

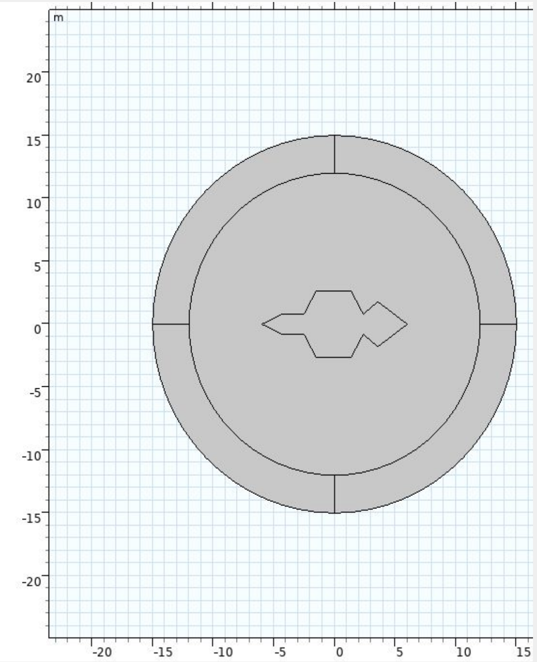


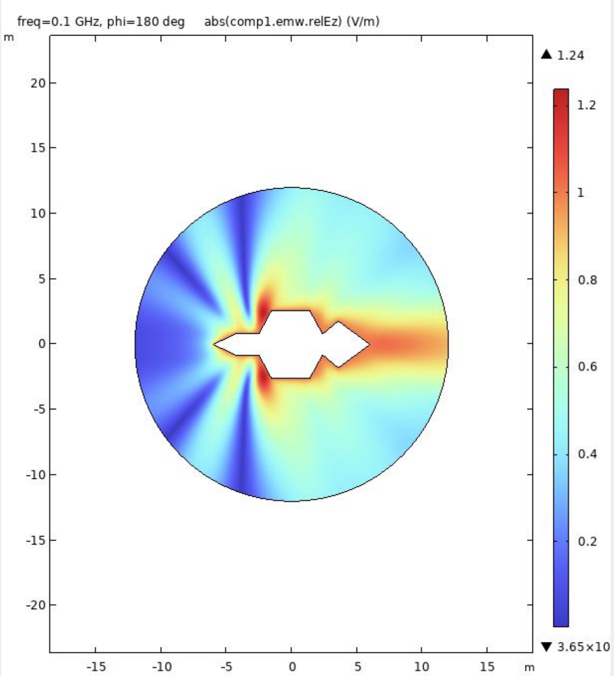


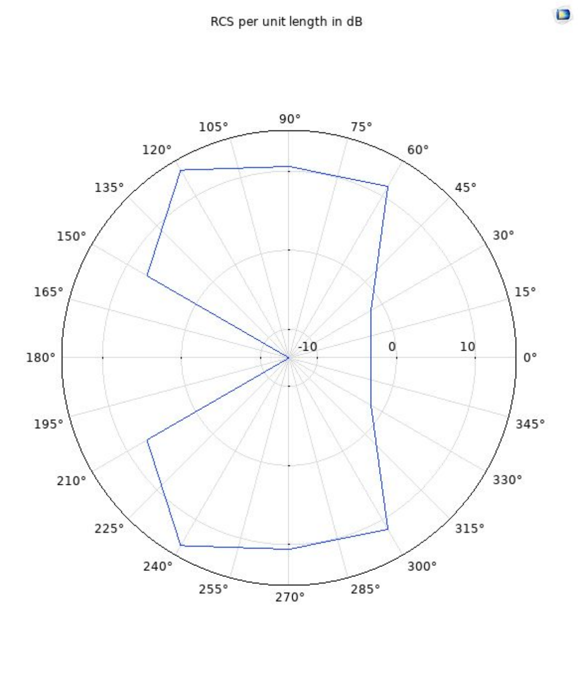


Method 2: parametric optimization

Variance in original shape

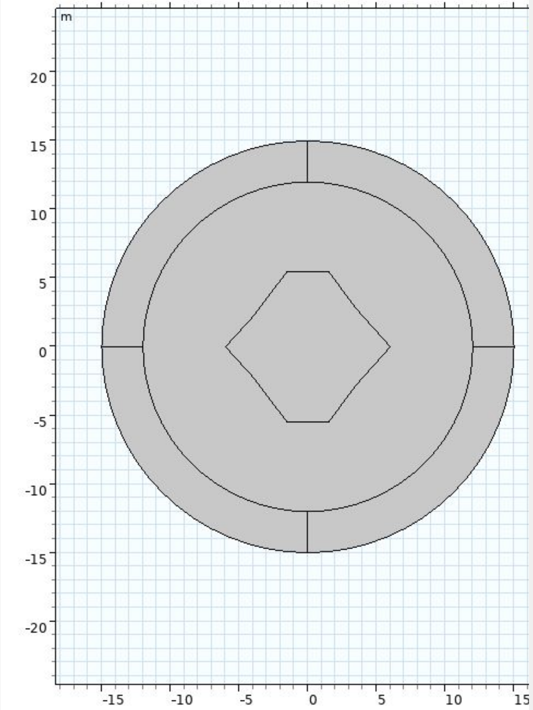


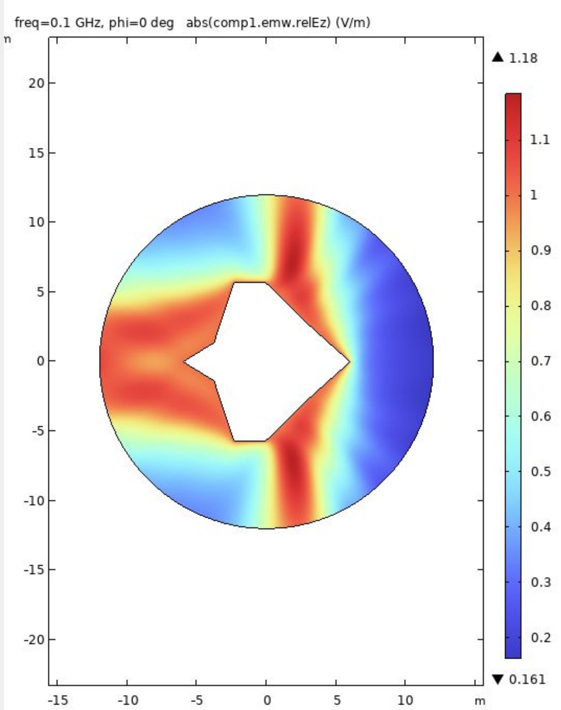


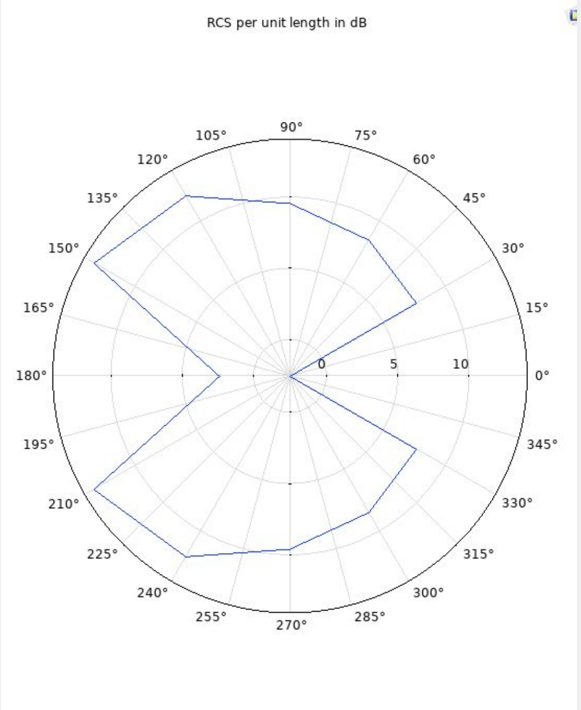


Method 2: parametric optimization

Variance in original shape

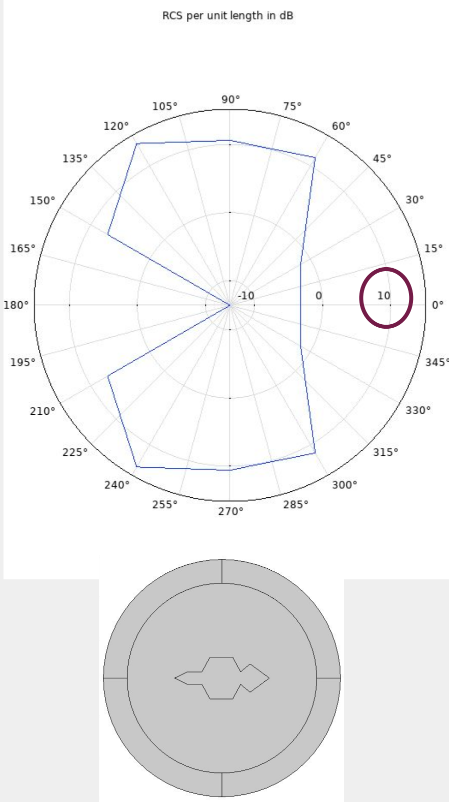


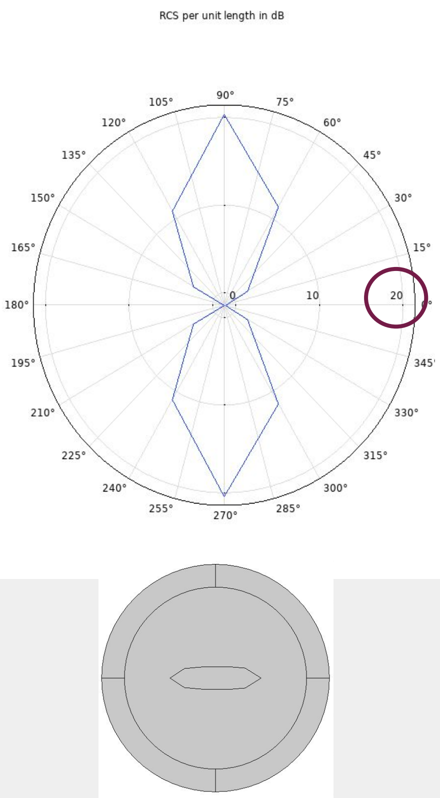


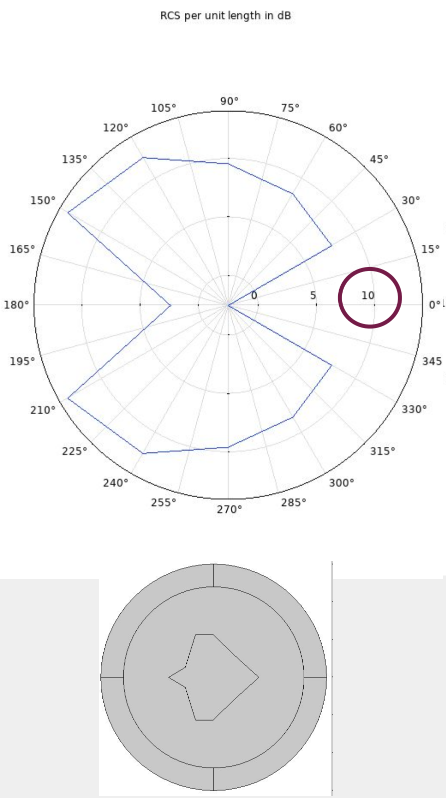


Method 2: parametric optimization

Compare: (reflected)

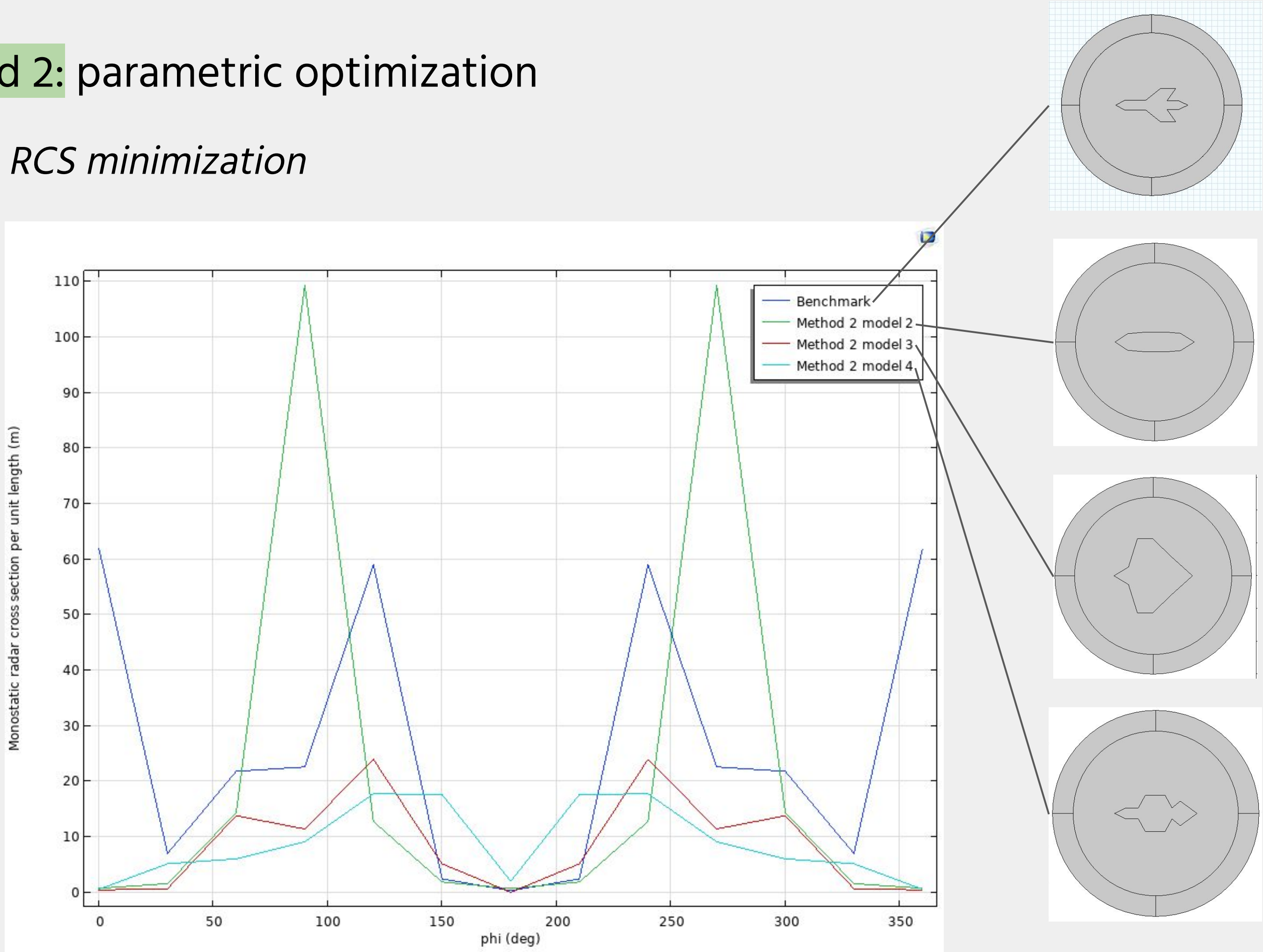






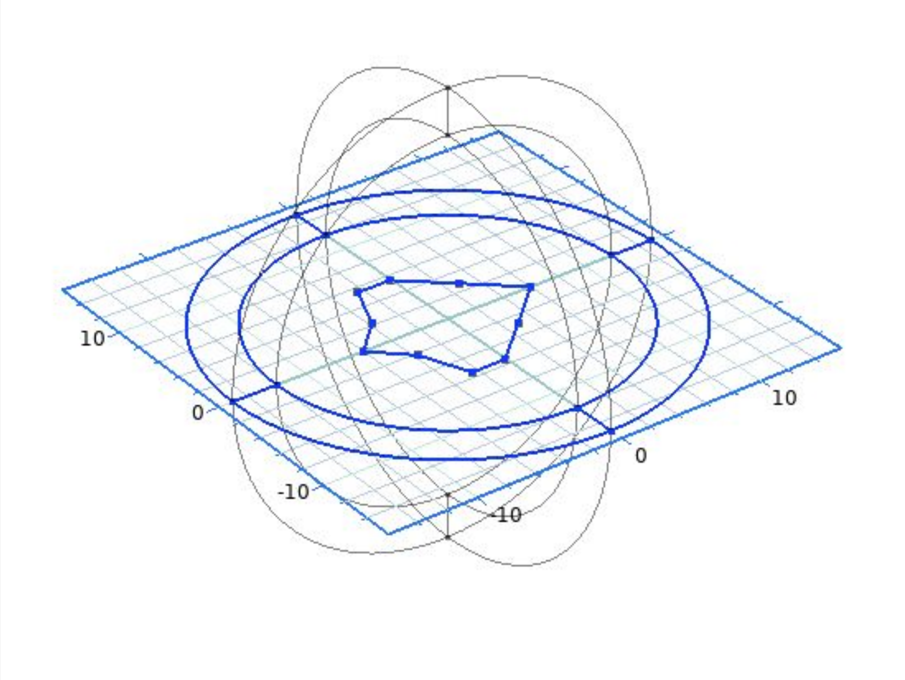
Method 2: parametric optimization

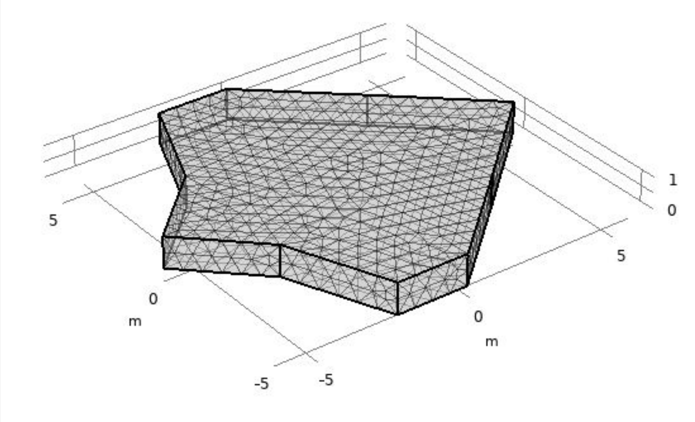
Results: RCS minimization

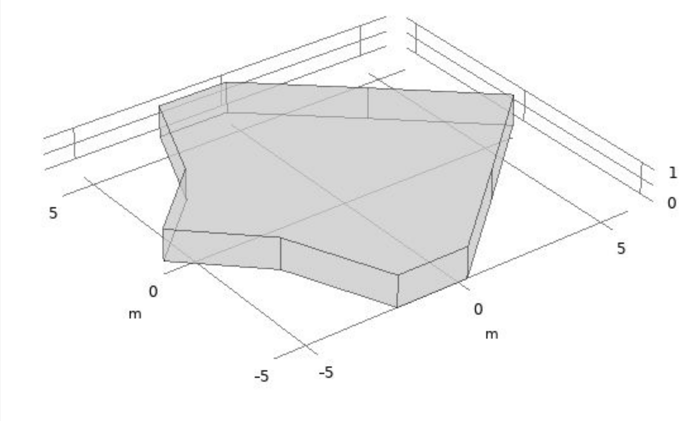


Method 2: parametric optimization

2D part → 3D part



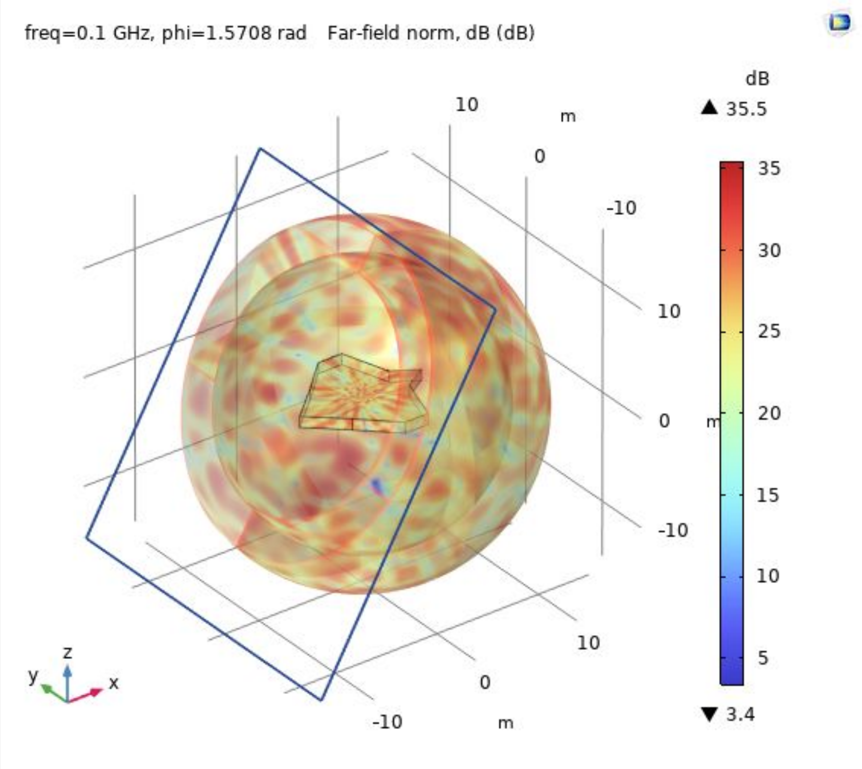


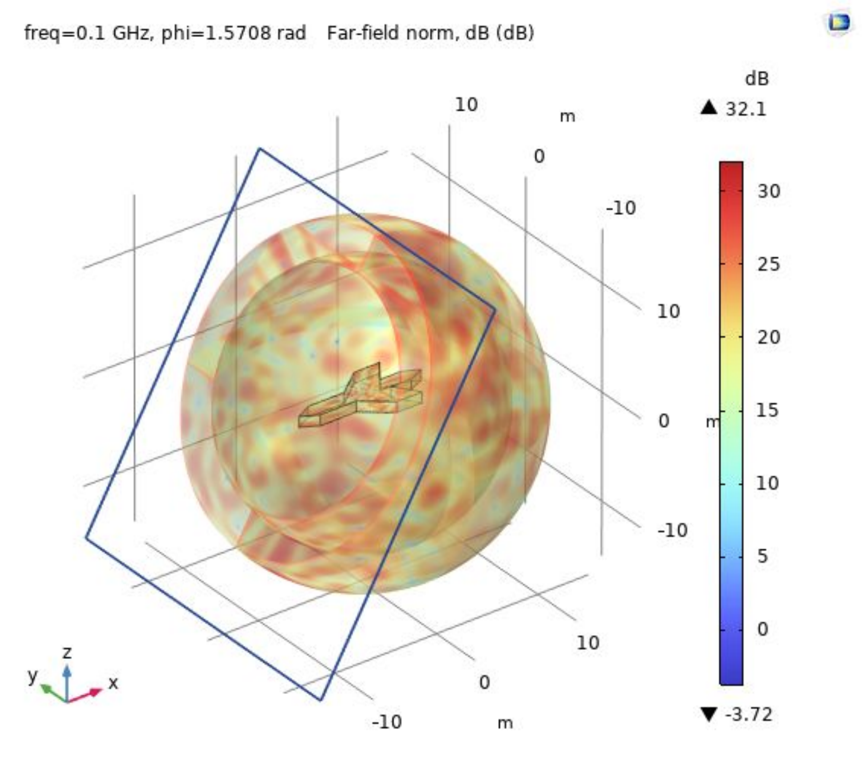


Method 2: parametric optimization

Results: far-field norm

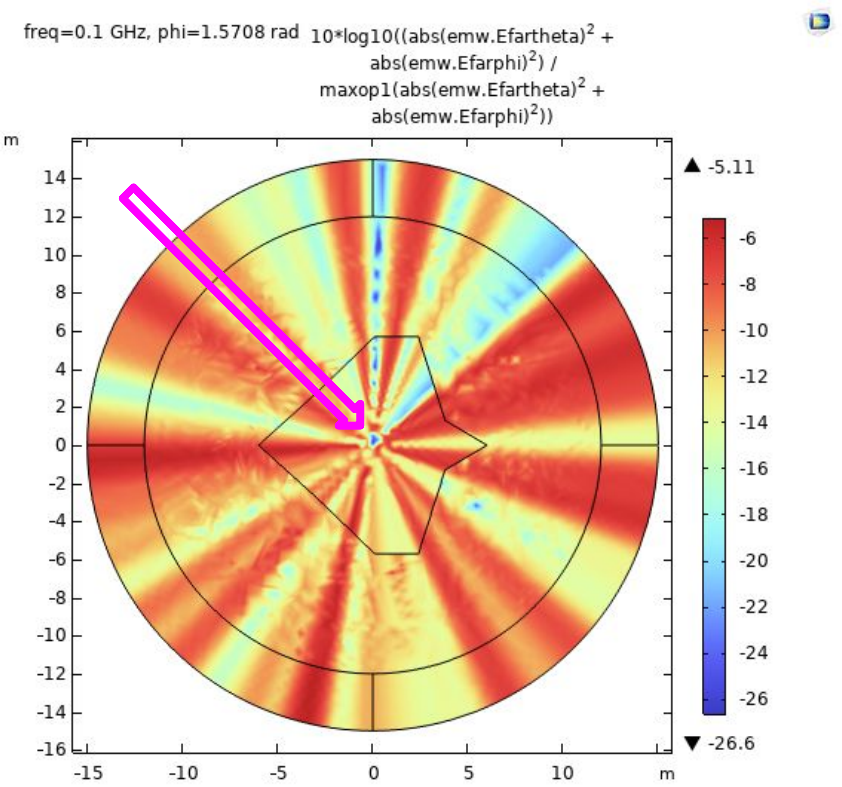
Compare:



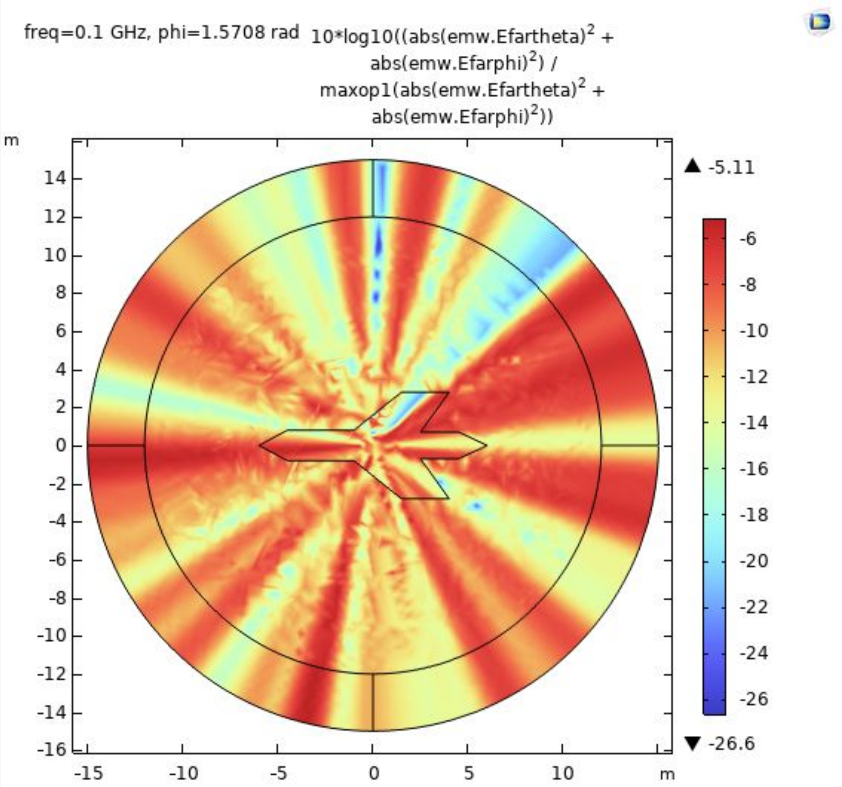


Method 2: parametric optimization

Results: relative electric field

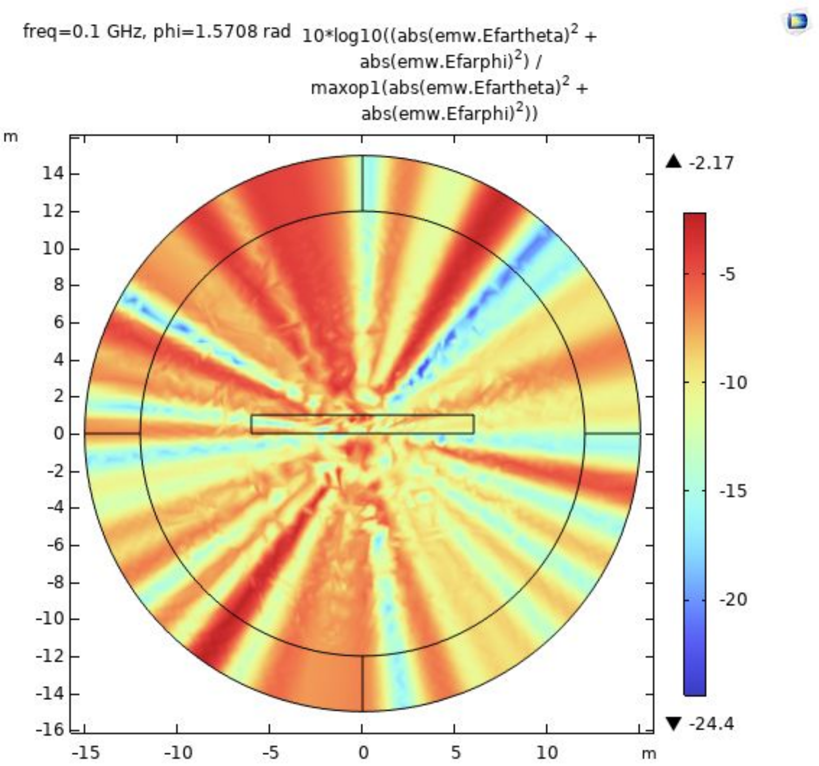


Compare:

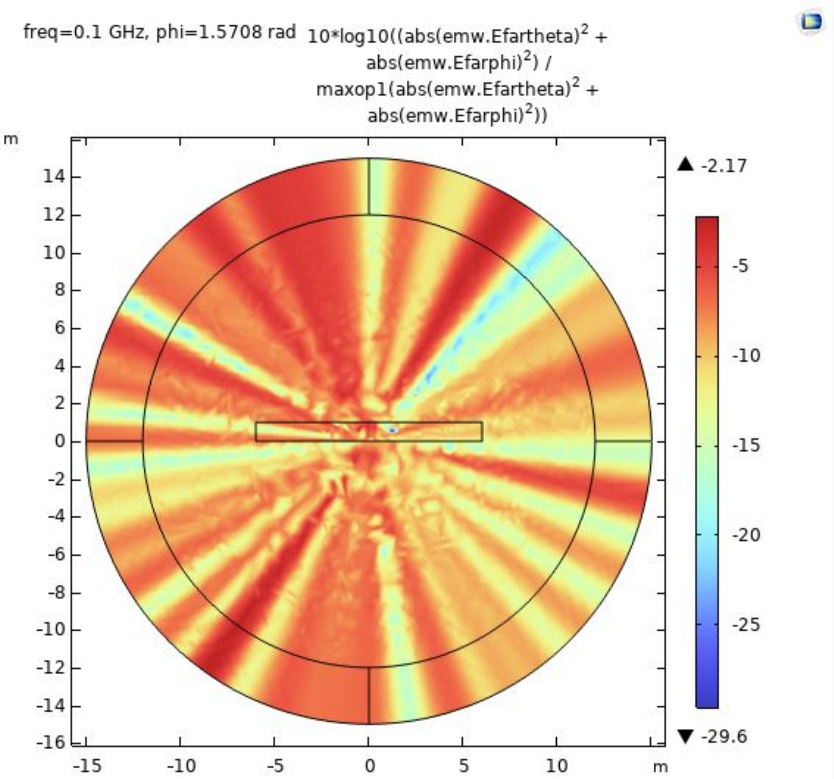


Method 2: parametric optimization

Results: relative electric field

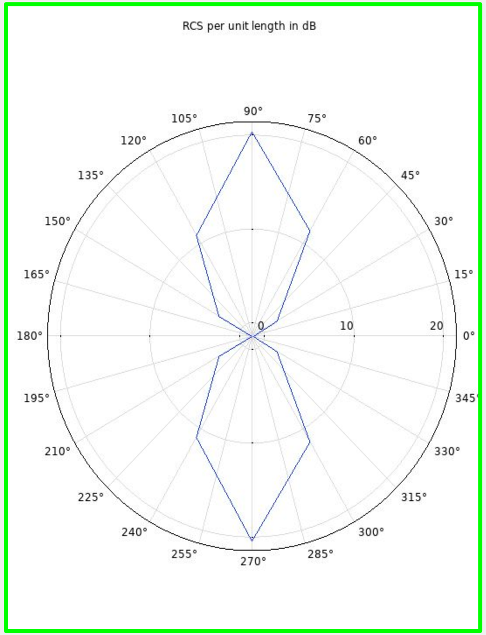


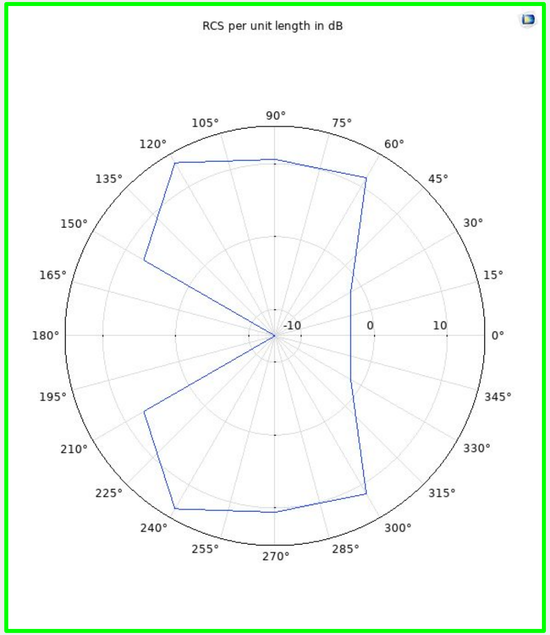
Compare:

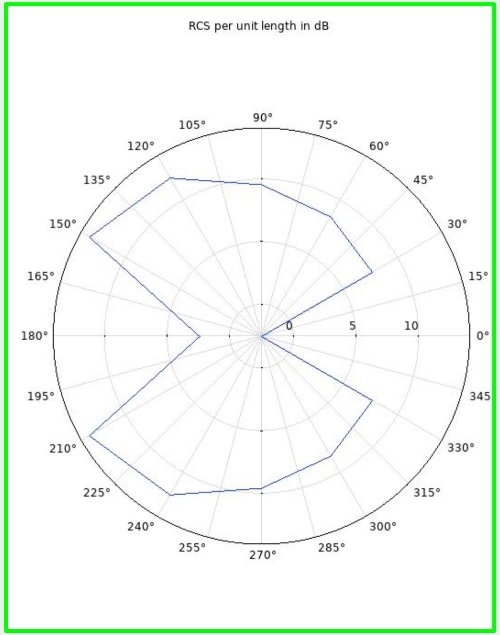


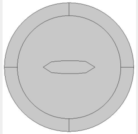
Method 3: advanced model

Taking from the most successful models...

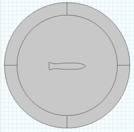




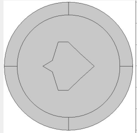




Method 2
model 2



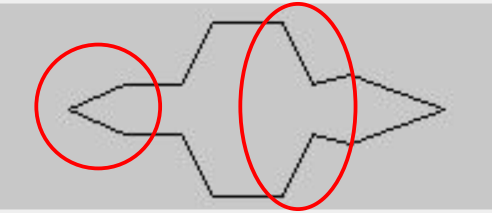
Method 2
model 3

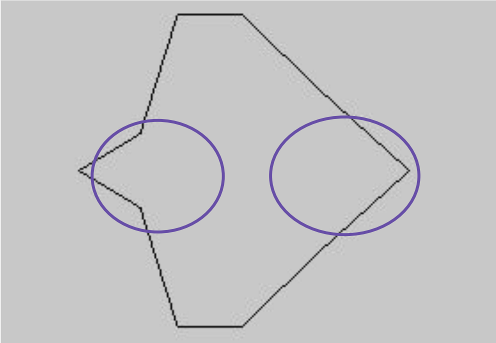


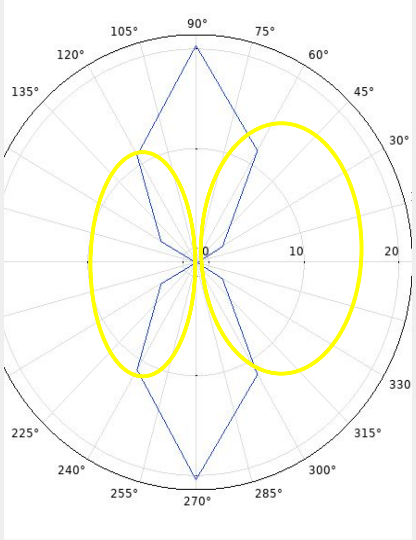
Method 2
model 4

Method 3: advanced model

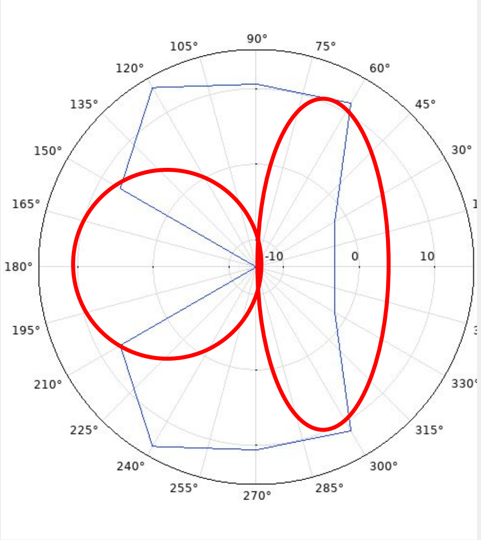
Taking from the most successful models…



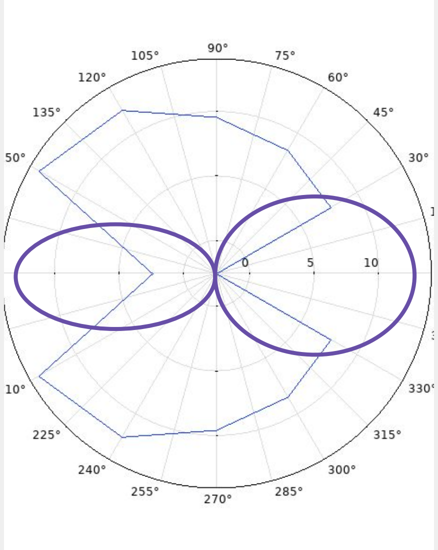




Method 2
model 2



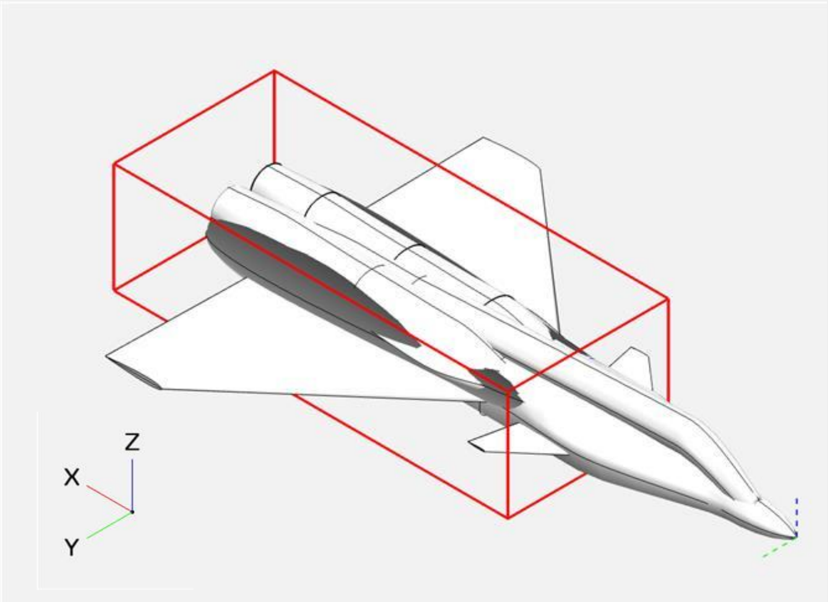
Method 2
model 3

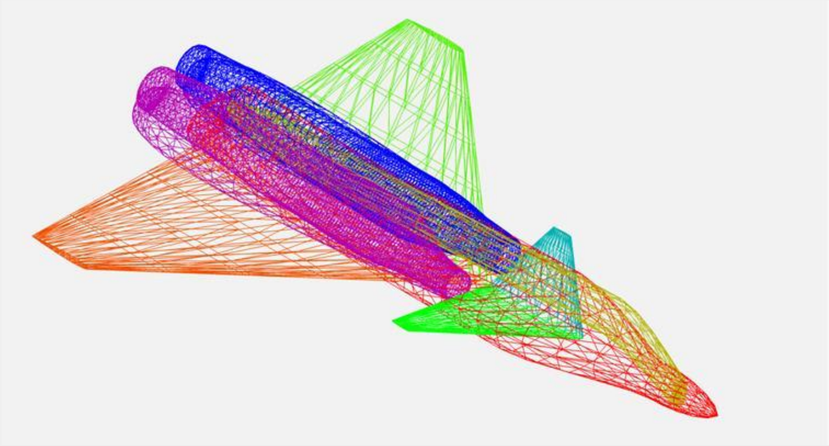


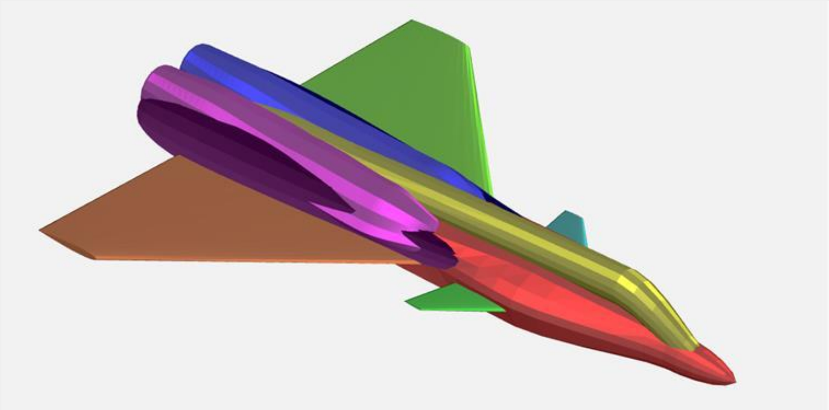
Method 2
model 4

Method 3: advanced 3D model

3D paramaterized plane in OpenVSP



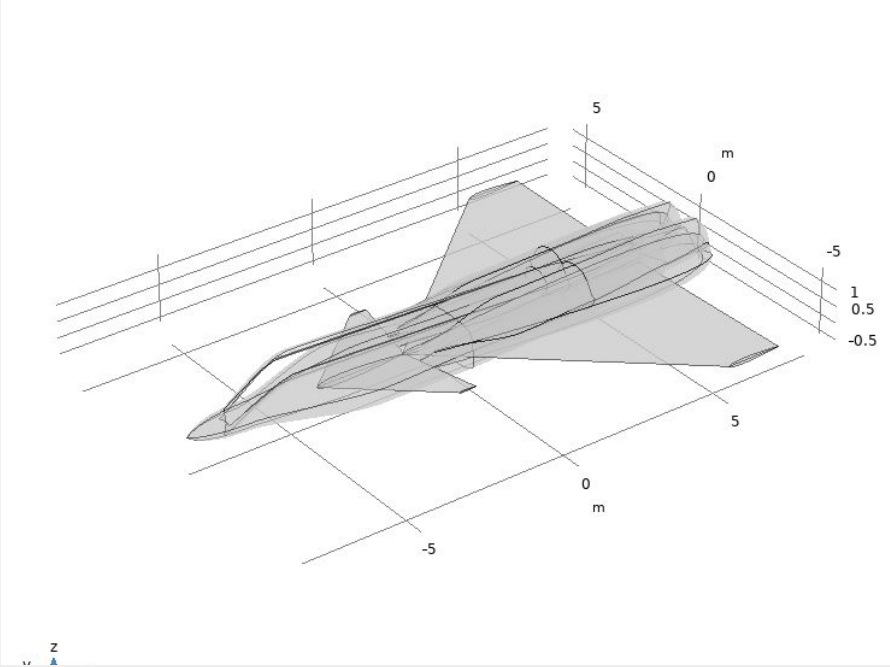


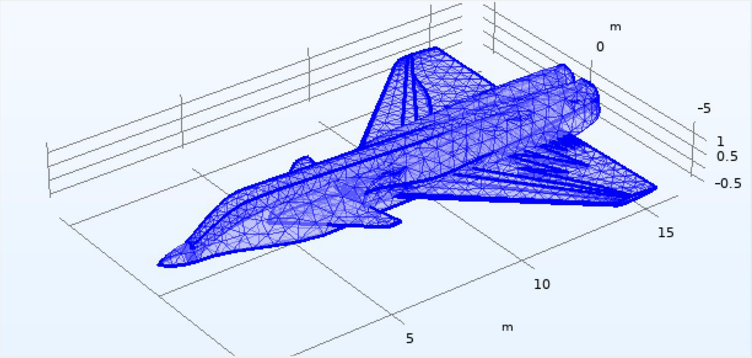


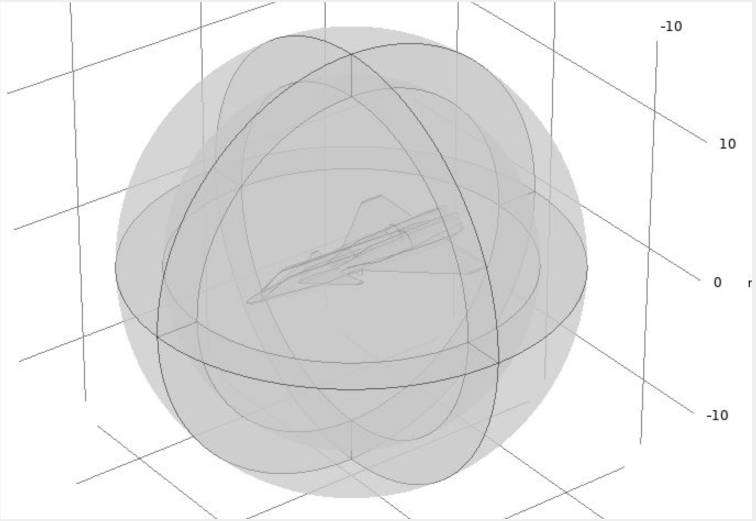
+ equipment considerations

Method 3: advanced 3D model

3D paramaterized plane in COMSOL

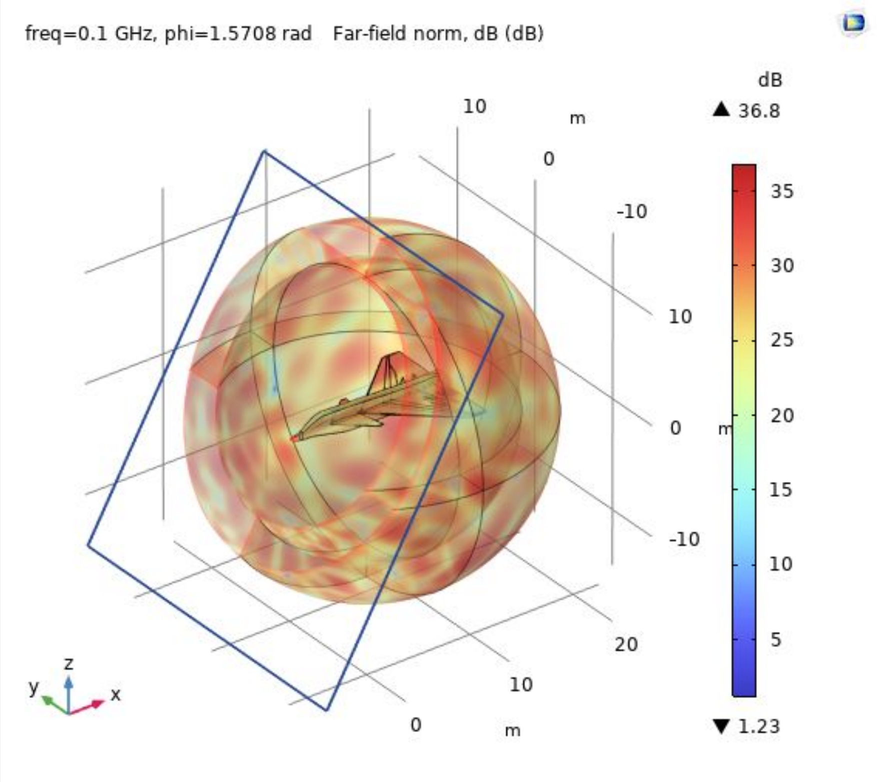




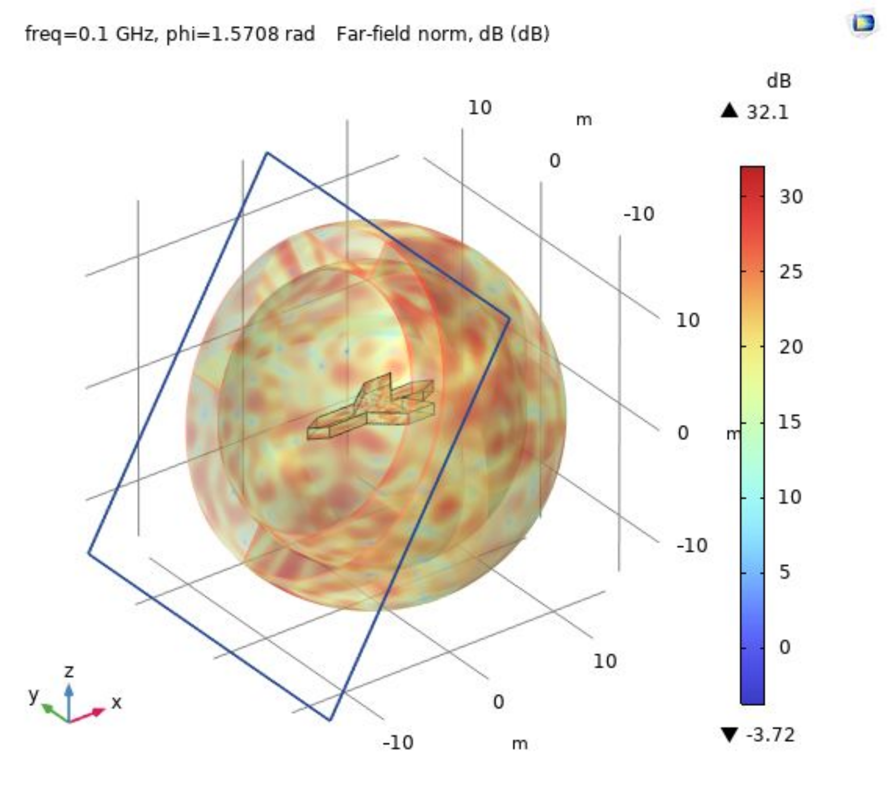


Method 3: advanced 3D model

Results: far-field norm

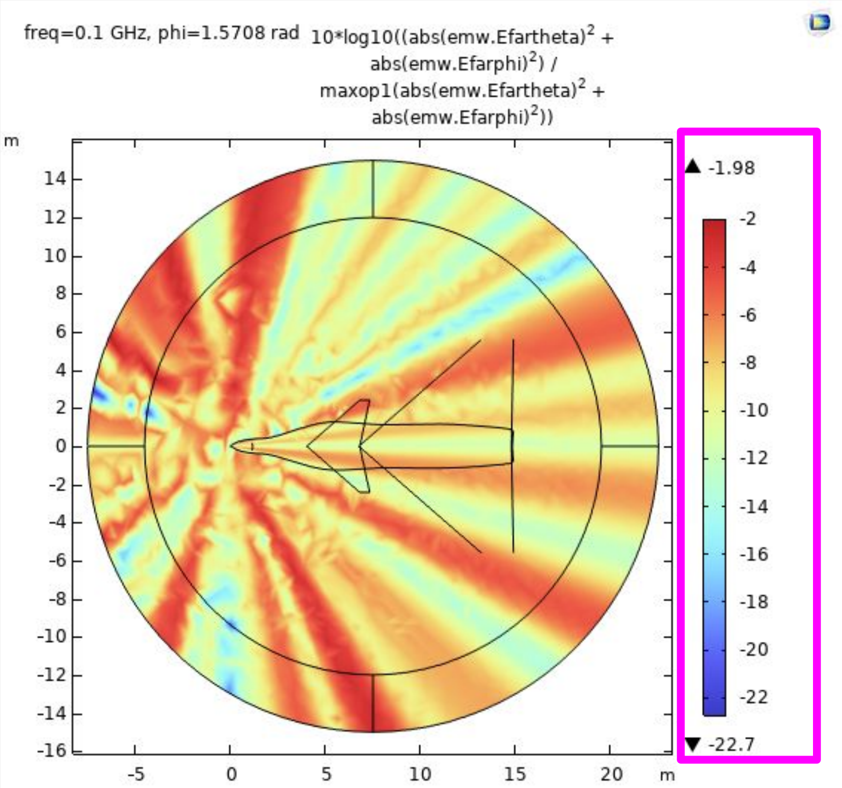


Compare:

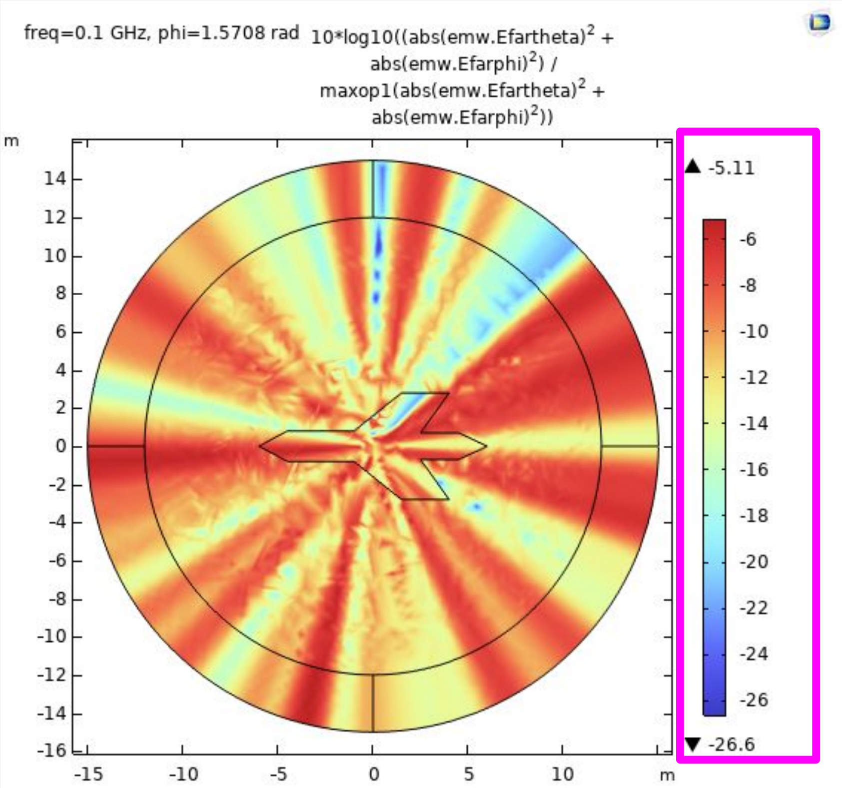


Method 3: advanced 3D model

Results: relative electric field

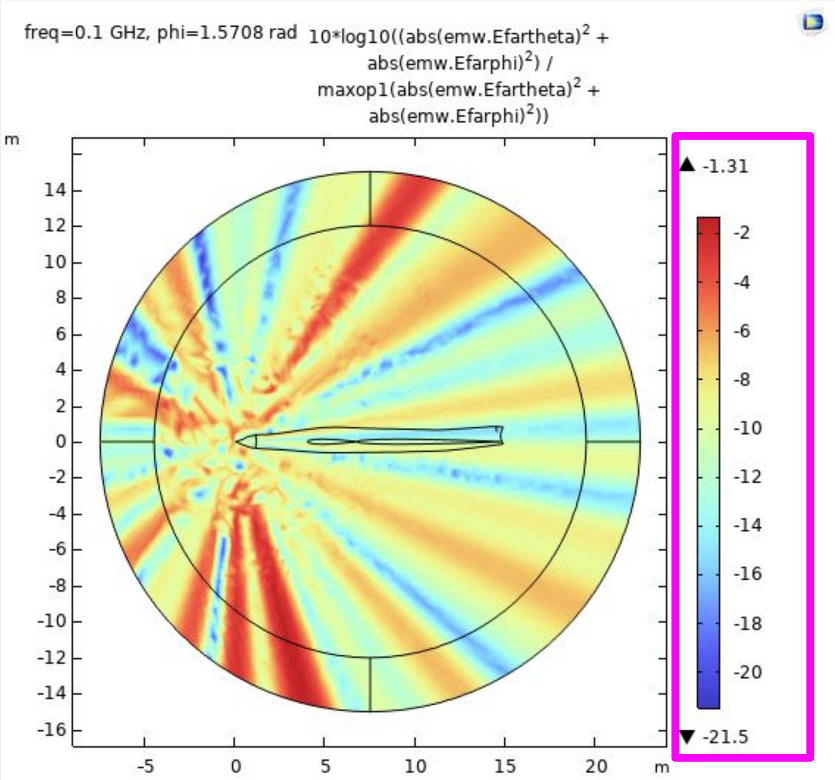


Compare:

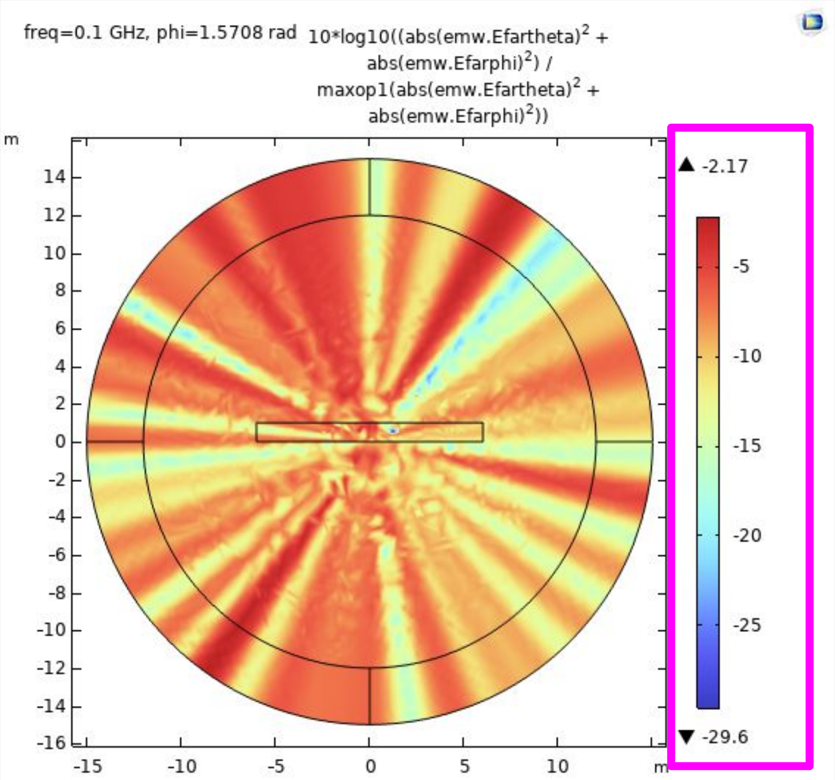


Method 3: advanced 3D model

Results: relative electric field



Compare:



Model Issues and Future Improvements

Current Issues	Future Improvements
Geometry Mesh - Difficult to refine for accurate results **Heavy Focus on 2D Computation** - Currently limited to single angular slice optimization of scattering **Complex Geometry Appear Identical** - Since geometry is constructed in 2D, many side profiles look identical	**Global 3D Optimization** - Allow for quantification over all relevant angles - Identify full surface scattering trends and secondary sources **Parametric Optimization** - Automate surface fine tuning - Allow for directional emphasis - Introduce material optimization